

Phage Therapy for MDR/XDR/PDR *Pseudomonas aeruginosa*: A Systematic Review and Meta-Analysis of Proportions

[Author Name(s)]*

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Abstract

Background: MDR, XDR and pandrug-resistant *Pseudomonas aeruginosa* is a WHO priority pathogen with few options; phage therapy has expanded without stratified synthesis. Methods: PRISMA 2020 meta-analysis of proportions (binomial-normal GLMM), stratified by resistance class, difficult-to-treat resistance, route and modality. Results: 40 pooling-eligible arms (33 studies, 91 patients) gave clinical success 78.7% (95% CI 68.0%–86.6%), a proportion of the published record, not of treated patients. 13 of 48 cells met threshold, including MDR 75.0% (95% CI 58.1%–86.6%). The highest estimate in the review, 88.5%, falls in the stratum defined by *absent* resistance documentation. XDR, PDR and every specific route stayed below threshold; modality could not be stratified, no arm being antibiotic monotherapy. τ^2 is zero in 11 cells, so intervals carry no between-arm variance. Conclusions: These proportions should not inform treatment decisions; the usable output is a map of what the published record cannot support.

Keywords: Bacteriophages; *Pseudomonas aeruginosa*; Drug Resistance, Multiple, Bacterial; Phage Therapy; Systematic Reviews as Topic; Meta-Analysis as Topic

*Universidad Católica de Cuenca, Cuenca, Ecuador. Corresponding author: [email address TBD]. Full author list, affiliations, and contribution statement (per ICMJE criteria) to be finalized before journal submission — no specific authorship has been established as of this draft.

1 Introduction

Pseudomonas aeruginosa appears on the World Health Organization’s priority list of antibiotic-resistant pathogens because of its capacity to accumulate resistance to nearly every class of antipseudomonal agent (Tacconelli et al., 2018). That framing needs one correction, and it cuts against rather than for the urgency of this review’s question: the 2024 update to the priority list *downgraded* carbapenem-resistant *P. aeruginosa* from Critical to High (World Health Organization, 2024). We note it explicitly rather than quietly citing the superseded 2018 list. The organism remains a High-priority pathogen for which salvage options are genuinely scarce, which is sufficient motivation; overstating its rank is not necessary and would not survive review.

A second correction is owed to the premise itself. It is no longer accurate to say the antipseudomonal pipeline has stalled: ceftolozane–tazobactam, ceftazidime–avibactam, imipenem–relebactam and cefiderocol all reached clinical use in the last decade and between them recovered activity against many isolates that older definitions call multidrug-resistant. The scarcity this review addresses is therefore not an absence of new agents but the narrower and harder situation those agents leave behind — the patient in whom they have also failed, or cannot be used. That distinction has a formal counterpart. The Magiorakos categories (MDR/XDR/PDR) count how many antimicrobial *categories* an isolate resists, a measure the new agents have made less decision-relevant; Kadri et al. (2018) instead define *difficult-to-treat resistance* (DTR) as non-susceptibility to all *first-line* agents — the carbapenem, extended-spectrum cephalosporin, β -lactam/ β -lactamase-inhibitor and fluoroquinolone categories — which is the point at which a clinician is forced onto reserve therapy. The reserve agents themselves, which Kadri et al. name as aminoglycosides, polymyxins and tigecycline, do *not* enter the determination: an isolate susceptible only to colistin is DTR precisely because colistin is what is left. DTR, not MDR, is the category current guidance uses for treatment decisions in *P. aeruginosa*. We retain the Magiorakos axis as primary because it is what the published literature reports, and we carry DTR as a separate stratum precisely

to show what that reporting practice costs (Sections 2.2 and 3.4). Once an isolate meets criteria for multidrug resistance (MDR), extensive drug resistance (XDR), or pandrug resistance (PDR) under the Magiorakos framework (Magiorakos et al., 2012), treatment options narrow quickly. The antipseudomonal development pipeline has not kept pace, and bacteriophage therapy—administered alone or alongside antibiotics—has moved from isolated compassionate-use reports toward a small but growing set of early-phase trials as a salvage option for infections that no longer respond to conventional regimens.

This evidence base has already been synthesized many times. Since El Haddad et al. (2019) reviewed phage therapy against ESKAPE pathogens in 2019, at least eight further systematic reviews, meta-analyses, and scoping reviews of human phage therapy have appeared, including a large multi-author synthesis in *Lancet Infectious Diseases* (Uyttendaele et al., 2022) and, most recently, an individual-patient-data meta-analysis of 130 studies spanning multiple pathogens (Liu, Thong, et al., 2025). None restricts its pooling to *P. aeruginosa* specifically, and none stratifies pooled outcomes by resistance category or by route of phage administration—the two variables clinicians most need resolved when choosing a regimen for a given patient and infection site.

Our original aim was narrower still: to compare, within the same studies, phage-antibiotic combination therapy against antibiotic monotherapy for MDR/XDR/PDR *P. aeruginosa* and to estimate a pooled effect size for clinical success. A systematic search of PubMed/MEDLINE and ClinicalTrials.gov found no study reporting both a genuine antibiotic-monotherapy arm and a phage-antibiotic combination arm in a *P. aeruginosa*-specific, MDR/XDR/PDR-defined population, so this comparison could not be constructed under any available within-study design. We therefore reframed the review around pooled proportions rather than a paired effect estimate: the proportion of patients achieving clinical success and the proportion experiencing a safety event across the single-arm and comparative studies that do exist, with pre-specified stratification by resistance category (MDR, XDR, or PDR), route of administration, and treatment modality (antibiotic monotherapy vs. phage-antibiotic com-

bination).

Specifically, among patients with MDR, XDR, or PDR *P. aeruginosa* infections treated with bacteriophage therapy, we asked: what proportion achieve clinical success, what proportion experience an adverse event, and do these proportions differ by resistance category, administration route, or treatment modality?

We found the current evidence base able to answer only part of the stratified question it was designed to ask. Pooling across all included studies yielded a clinical-success proportion of 78.7% (95% CI 68.0%–86.6%), directionally consistent with the high success rates reported by prior, non-stratified syntheses—but this figure comes from a compassionate-use population selected for refractory infection after antibiotic failure and should not be read as evidence that phage therapy outperforms, or underperforms, any comparator. Stratification succeeded, but only just, for one resistance class: MDR isolates specifically now have a pooled estimate (75.0% (95% CI 58.1%–86.6%) clinical success). We present this as a single-cohort characterization rather than a headline achievement: it draws two-thirds of its patients from one cohort and clears the pre-specified evidentiary threshold only under author-reported resistance labels—restricted to independently re-derived classifications, the MDR cell collapses to three studies and four patients and cannot be pooled at all. XDR, PDR, every specific administration route, and modality all fell short of the number of studies and patients needed for a stable pooled estimate, and despite a targeted search, no eligible antibiotic-monotherapy arm was identified—so the MDR-versus-XDR comparison that motivated this review remains unanswered even though MDR alone is not. Documenting exactly which parts of this stratified question the current evidence can and cannot answer, rather than forcing an estimate the data cannot support or dismissing the one part that succeeded, is the contribution of this review.

2 Methods

This review follows the PRISMA 2020 reporting standard (Page et al., 2021) and the methodological guidance of the Cochrane Handbook (Higgins et al., 2023), adapted throughout for a synthesis of proportions rather than a synthesis of comparative treatment effects, for the reasons given in the Introduction. Every design choice below—eligibility, search, extraction, risk-of-bias assessment, and the pooling model itself—was fixed before formal data extraction began, and is reported here exactly as specified, including the places where the evidence base did not cooperate with the plan and the places where this review’s own execution fell short of its own plan.

2.1 Protocol and Registration

This review was not registered in PROSPERO before literature screening or data extraction began. The design pivoted once during discovery, from a comparative-effect meta-analysis (phage-antibiotic combination versus antibiotic monotherapy) to the stratified meta-analysis of proportions reported here, after two independent search rounds found zero to one qualifying dual-arm comparative studies for MDR/XDR *P. aeruginosa*. That pivot needed to happen before a protocol could be written that the team would not have to immediately amend. That scoping work included four rounds of literature searching (Section 2.3) and one informal, single-reviewer screening pass of a 201-document local PDF corpus, conducted against this review’s PICO criteria but without the dual-independent review PROSPERO registration presumes. PROSPERO’s own policy does not prohibit preliminary scoping before registration; it prohibits registering a protocol once data extraction and synthesis are already substantively underway. By that standard, registration remained available through the design-pivot and literature-search stages of this project, and did not occur. Data extraction and the statistical synthesis reported in Results proceeded before a protocol was ever submitted – a genuine deviation from prospective registration – and that extraction

itself was conducted by a single reviewer rather than the dual-independent process a registered protocol would specify (Sections 2.4 and 2.5). The full scoping timeline is as follows: two PubMed/ClinicalTrials.gov search rounds and a targeted third and fourth round (Section 2.3), one ad hoc single-reviewer PDF screen, and the resulting design pivot, all preceded any protocol filing, and no protocol was filed before extraction and pooling were carried out. We recommend retrospective PROSPERO registration disclosing this exact sequence before this review is submitted for publication.

2.2 Eligibility Criteria

Eligibility follows a PICO framework, broadened partway through discovery once the design pivoted from a comparative-effect estimand to a proportion estimand (Section 2.1): under the original comparative-only design, a study needed both a phage-containing arm and a within-study antibiotic-monotherapy arm to qualify, a requirement that essentially no study in this literature satisfies. Under the current design, single-arm and comparative studies are equally eligible, because every outcome is computed at the study-arm level rather than as a paired comparison.

- **Population.** Patients with an infection attributed to multidrug-resistant (MDR), extensively drug-resistant (XDR), or pandrug-resistant (PDR) *Pseudomonas aeruginosa*, classified per the Magiorakos et al. (2012) criteria where the primary study reports sufficient susceptibility data, and per the study’s own MDR/XDR/PDR label otherwise (Section 2.5). Three cases must be distinguished, and this review treats them differently.

First, where a study reports susceptibility data that *positively fail* the Magiorakos threshold, the arm does not meet the population criterion and is excluded, whatever the study’s own descriptive language. Second, and consequentially, an arm is excluded where its source population was *recruited without any resistance requirement* — that

is, where a trial’s own eligibility criteria admit patients on the basis of *P. aeruginosa* infection or chronic airway colonization alone. For such arms the missing resistance label is not a reporting gap: the population demonstrably was not selected for resistance, and pooling it would place patients who may be fully susceptible inside a review of resistant infection. Third, where an individual compassionate-use or salvage report simply does not *report* a susceptibility panel, the arm is retained and coded *not classifiable*, because absence of a reported panel is not evidence of susceptibility and the population is the refractory, antibiotic-failed population this review targets.

The distinction between the second and third cases is the load-bearing one, and we state it plainly: a missing susceptibility panel in a salvage case report is a documentation gap, whereas a missing resistance criterion in a trial protocol is a different population. Applying the second rule consistently removes every randomized trial from the synthesis — PhagoBurn, whose burn-wound population required no MDR status; and the BX004-A and SWARM-P.a./AP-PA02 cystic-fibrosis trials, which enrolled on chronic pulmonary colonization — together with three further completed trials identified in a registry sweep (Section 2.3). We accept that cost deliberately. Those trials were never estimating this review’s estimand, and they are more informative as external comparative evidence (Section 4.2) than as single-arm proportions inside a synthesis whose population they do not share. What remains is a corpus of compassionate-use and salvage reports in which resistance is positively established for 28 of 40 arms; the other 12 arms, carrying 26 patients or 29% of the corpus, report no resistance information at all. We do not minimise that residual: it is the largest single concession this eligibility rule makes, and a sensitivity analysis reports what excluding those arms does to every pooled estimate (Section 3.6).

- **De-duplication against the Belgian consortium cohort.** Pirnay et al. (2024) reports the first 100 *consecutive* cases facilitated by the Queen Astrid Military Hospital consortium, and its three resistance-stratified arms are in this corpus. Any separately

published case report describing a patient inside that roster would therefore be counted twice. Van Nieuwenhuyse et al. (2022) was identified as exactly such a duplicate during screening, which established the risk as real rather than theoretical, so a systematic crosswalk was run against the roster in that paper’s Supplementary Table 1. **That crosswalk has a material limitation we state rather than bury: the roster carries no country column and no treatment-year column**, so matching could only use infection type, species, resistance profile, bacteriophage product and outcome pattern — not provenance. Fifty of the 100 roster cases involve *P. aeruginosa*. Two of this review’s case reports matched closely enough that duplication could not be excluded, and under a conservative rule the case report is dropped and the cohort entry retained: Ferry, Kolenda, Laurent, et al. (2022) against roster case 79 (the only chronic spondylodiscitis in the roster, same species, same Belgian phages, and matching this review’s own extraction of that case as neither eradicated nor surviving), and Racenis et al.’s left-ventricular-assist-device case against roster case 56 (same indication, species, resistance class and QAMH phage source). The Ferry match is the weaker of the two: the roster codes case 79 XDR while the case report derives PDR from published MICs. Three further candidates — Racenis et al.’s femoral case, Blasco et al. (2023) and Tkhilaishvili et al. (2020) — were examined and *retained*, because no roster case matched on resistance class. Because these matches rest on clinical rather than provenance fields, they are a judgement and not a determination; Section 3.6 reports what the pooled estimates become if both arms are put back.

- **Resistance classification and the DTR axis.** Arms are classified on the Magiorakos et al. (2012) axis (MDR/XDR/PDR/not classifiable) as above. A *fourth* stratification axis records difficult-to-treat resistance (Kadri et al., 2018), defined as intermediate or resistant status to all reported first-line agents — carbapenems, extended-spectrum cephalosporins, β -lactam/ β -lactamase-inhibitor combinations and fluoroquinolones. Aminoglycosides, polymyxins and tigecycline are reserve agents in

Kadri et al.’s scheme and are excluded from the determination, as are agents postdating it (ceftazidime-avibactam, ceftolozane-tazobactam, cefiderocol), which Kadri et al. note a future revision would need to address. Two consequences follow and both are applied here: an isolate susceptible only to reserve agents is DTR-*positive*, and a single documented susceptible first-line agent is sufficient to establish DTR-*negative*. Earlier rounds of this review required non-susceptibility to *all* β -lactams, which is stricter than Kadri et al. and under which no arm could be shown negative; six arms were re-adjudicated when the criterion was corrected. DTR is not a fifth Magiorakos level and is not nested inside that axis: it asks a different question — whether every first-line agent is lost — so an isolate can be MDR and DTR, or MDR and *not* DTR while cefepime remains active. It is therefore stratified in parallel. Coding is deliberately conservative: an arm is DTR-positive only where PDR status *entails* it, since non-susceptibility to all agents in all categories necessarily includes all β -lactams and both fluoroquinolones; every other arm is recorded as *not derivable* rather than assumed negative, because XDR permits retained susceptibility in up to two categories and MDR permits far more. Magiorakos is retained as the primary axis for one reason, stated plainly: it is what this literature publishes. The DTR axis is carried not because we expect to estimate it but because the pattern of its failure is itself the finding (Section 3.4).

- **Intervention.** Bacteriophage therapy, administered as monotherapy or adjunctively with antibiotics, by any route. One qualification is load-bearing and was previously applied without being stated, which we correct here. A phage preparation that cannot lyse the patient’s organism is not a dose of the intervention this review is estimating, so an arm is excluded where the administered preparation was *never* documented active in vitro against the *P. aeruginosa* isolate. The rule is deliberately narrow — *never* active, not *initially* inactive — and the two cases that turn on it show why the distinction is not a convenience. Onsea et al. (2019) is excluded: its XDR arm received a cocktail

to which the isolate was non-susceptible, and no active preparation followed, so the arm reports the natural history of a treated wound rather than the effect of a phage. Patient 2 of Zaldastanishvili et al. (2021) is *retained* although her strain was resistant to the pre-made Pyo and Intesti products from the first culture, because five successive phages were then prepared against that strain and it is those she received. Excluding her would penalise a study for reporting the personalisation step that this field regards as best practice, while excluding Onsea et al. turns on the absence of any active agent at all.

We record what this criterion costs. Beyond the two adjudications above, this review does *not* extract the phage preparation itself — not the product or its constituent phages, not titre, not dose, not schedule, not duration, and not whether the preparation was personalised or pre-made. The pooled proportions therefore treat a single-phage personalised preparation and a multi-phage commercial cocktail as the same exposure, which they are not. This is a genuine limitation of the review and not only of the literature, and it is the reason no dose- or product-stratified analysis appears below; the corresponding gap in the primary sources — how rarely they report whether the phage was active on the organism at all — is quantified in Section 3.4 and carried into the certainty assessment as indirectness.

- **Comparator.** None required. Where a genuine within-study comparator exists (an antibiotic-only or placebo arm), it is retained and reported, but its presence is not an eligibility condition.
- **Outcomes.** Clinical success and any adverse event (primary); microbiological eradication, mortality, length of hospitalization, and resistance emergence during treatment (secondary), each defined per the primary study’s own reporting given the expected heterogeneity in how “success” is defined across this literature. Because each study defines clinical success on its own terms *and* across non-comparable infection syndromes —

biliary, cystic-fibrosis lung, prosthetic-joint, vascular-graft, urinary-tract, bacteremia, and osteomyelitis among those represented in this corpus — the pooled clinical-success proportion aggregates outcomes that are not the same clinical construct, and should be read as a rough descriptive summary rather than a harmonized effect estimate. A syndrome-harmonized success definition was not achievable given the heterogeneous reporting of the primary literature, and this constraint is carried forward as a limitation (Section 4.4).

- **Study designs.** Randomized controlled trials, prospective and retrospective cohorts, case series, and case reports, including compassionate-use reports.

A study-arm was excluded if its causative pathogen could not be separated from a mixed-pathogen cohort (coded `mixed-not-separable` in the extraction schema; Section 2.5), regardless of how well the rest of the study otherwise matched. The primary electronic search (Section 2.3) was restricted to records published from 2016 onward across all three core databases. This 2016 cutoff was chosen as a pre-specified window that comfortably predates the modern clinical phage-therapy case-report era for *P. aeruginosa* — which begins in earnest around 2018–2019 (the earliest included study is Duplessis et al., 2018) — while still excluding the sparse, mostly single-institution reports from before the current compassionate-use framework. A targeted screen of the 2016–2018 window (Section 2.3) confirmed that the pre-2019 literature for this pathogen is almost entirely preclinical (phage isolation, characterization, biofilm and animal-model work); the only extractable human clinical studies in that window are Duplessis et al. (2018) and the PhagoBurn trial (Jault et al., 2019), both of which are included. A small number of still-older studies were also identified through citation-chasing of other reviews’ reference lists (Section 2.3) and screened against the same eligibility criteria irrespective of publication date; one such study (Rhoads et al., 2009) failed eligibility on grounds unrelated to date and is retained only as background context. The discovery-phase search separately confirmed that, contrary to the project’s initial premise, at least eight further systematic reviews, meta-analyses, or scoping reviews of human phage therapy

have appeared since El Haddad et al. (2019), including a broader synthesis in a comparably high-impact journal (Uyttebroek et al., 2022) and a 130-study individual-patient-data meta-analysis published only months before this review’s search began (Liu, Thong, et al., 2025).

2.3 Information Sources and Search Strategy

The primary electronic search was run directly against PubMed/MEDLINE using the NCBI E-Utilities API (`esearch`, `esummary`, `efetch`), and against ClinicalTrials.gov using its structured API (v2), across four documented rounds between 2026-07-18 and 2026-07-20. A representative query, run to test the central negative claim of this review (that essentially no dual-arm comparative study of phage-antibiotic combination versus antibiotic monotherapy exists for MDR/XDR *P. aeruginosa*), took the form:

```
("Pseudomonas aeruginosa"[tiab] OR "P. aeruginosa"[tiab]) AND ("phage"[tiab] OR
"bacteriophage"[tiab]) AND ("multidrug-resistant"[tiab] OR "MDR"[tiab] OR "extensively
drug-resistant"[tiab] OR "XDR"[tiab]) AND ("matched cohort"[tiab] OR "propensity
score"[tiab] OR "registry"[tiab] OR "retrospective comparative"[tiab] OR "case-control"[tiab])
```

This query, built directly from the terms a referee would be expected to demand, returned one hit, and that hit was a conference-abstract index record with no substantive connection to phage therapy: an effective true hit count of zero. Across all four rounds, PubMed queries of this kind returned 54 raw hits screened individually for the comparative-design question and 48 raw hits for the systematic-review/meta-analysis recheck; ClinicalTrials.gov structured queries returned 10 additional trial records screened individually. Round 3 added five previously uncatalogued comparative or randomized trials (Krakhotkin et al., 2025; Leitner et al., 2021; Karn et al., 2024; Rhoads et al., 2009; Stanley et al., 2025), found by citation-chasing the risk-of-bias tables in the largest existing phage-therapy meta-analysis’s supplementary dataset (Liu, Thong, et al., 2025) instead of a new keyword search. This is standard, low-risk PRISMA practice: only citation leads were imported at this stage

(Section 2.5 details the boundary between this citation-chasing use and any reuse of Liu et al.’s outcome data). Round 4 conducted a targeted, independent PubMed search for antibiotic-only cohort studies in MDR/XDR *P. aeruginosa*, unrelated to any phage-therapy keyword, specifically to check whether an external comparator population could substitute for the missing within-study monotherapy arm; that search returned 44 hits and confirmed that antibiotic-only outcome cohorts are abundant in the general infectious-disease literature, but none contains a phage arm and none is therefore usable as anything other than background context (Section 2.4).

We also queried WHO ICTRP and PROSPERO directly. Both attempts failed for the same reason: WHO ICTRP’s search portal is an ASP.NET postback-driven form, and PROSPERO’s search interface returns HTTP 403, and neither can be driven by a static, unauthenticated query in the environment available to this review team. A web-search-mediated check of both registries found no competing protocol matching this review’s exact population, intervention, and stratification design, but this constitutes a partial, not exhaustive, verification. A researcher with direct, authenticated access to both portals should re-run this check before this review’s registration (Section 2.1) is finalized.

The search rested on three databases — PubMed/MEDLINE, ClinicalTrials.gov, and Scopus, all searched natively — supplemented by citation-chasing, a standard and defensible base for a systematic review. To place PubMed/MEDLINE on the same topical footing as Scopus rather than leaving it represented only by the narrow comparative-design rounds above, a broad topical PubMed/MEDLINE search was run directly against the NCBI E-Utilities API on 2026-07-23, mirroring the Scopus scope: title/abstract terms for *Pseudomonas aeruginosa*, for phage or bacteriophage therapy, and for multidrug, extensive, pan-, or general drug resistance, restricted to the pre-specified 2016–2026 window:

```
("Pseudomonas aeruginosa"[tiab] OR "P. aeruginosa"[tiab]) AND ("phage"[tiab] OR
"bacteriophage"[tiab] OR "phage therapy"[tiab] OR "bacteriophage therapy"[tiab])
AND ("multidrug-resistant"[tiab] OR "MDR"[tiab] OR "extensively drug-resistant"[tiab]
OR "XDR"[tiab] OR "pandrug-resistant"[tiab] OR "PDR"[tiab] OR "drug-resistant"[tiab])
```

OR "drug resistance"[tiab])

This topical PubMed search returned 481 records over 2016–2026. Screened by document type and title against the same eligibility criteria, every eligible human clinical study it surfaced was already among those identified through the native Scopus search — consistent with Scopus indexing a superset of this topic’s clinical literature — so the topical PubMed search added no unique-new eligible study, and is documented here as a topical core-database search confirming the Scopus-identified corpus rather than extending it. A dedicated screen of the 2016–2018 sub-window (37 of the 481 records) confirmed it contains no extractable human clinical study beyond Duplessis et al. (2018) and PhagoBurn (Jault et al., 2019), both already included: the remaining pre-2019 records are preclinical (phage isolation, characterization, biofilm, and animal-model studies). The Scopus search was run directly against the native Scopus interface on the same date, exporting the full result set as an 860-record RIS file for screening. The query combined TITLE-ABS-KEY terms for *Pseudomonas aeruginosa*, for phage or bacteriophage therapy, and for multidrug, extensive, pan-, or general drug resistance. The RIS export executed used PUBYEAR > 2018; because the topical PubMed search above establishes that the 2016–2018 window holds no additional eligible clinical study for this pathogen, the corpus is unaffected by this bound, though a Scopus re-export from 2016 is recommended before final submission for a fully uniform per-database audit trail:

```
TITLE-ABS-KEY("Pseudomonas aeruginosa" AND (phage OR bacteriophage) AND (MDR OR  
XDR OR PDR OR "multidrug-resistant" OR "drug-resistant")) AND PUBYEAR > 2018
```

All 860 records were screened. The native Scopus search materially enlarged the corpus: it surfaced ten eligible human clinical *P. aeruginosa* phage-therapy studies the PubMed/MEDLINE and ClinicalTrials.gov rounds had not returned — Köhler et al. (2023), a single inhaled-phage MDR case, and Chan et al. (2025), a nine-patient nebulized-phage cystic-fibrosis cohort, together with eight further reports: Law et al. (2019), Hahn et al. (2023), Levêque et al. (2023), Duplessis et al. (2018), Denis et al. (2026), Malhotra et al. (2026), Yang et al. (2025),

and Li et al. (2025), an 88-year-old woman with an MDR biliary-tract infection treated by multi-route phage plus antibiotics (previously carried as a pending record, now resolved from two concordant secondary sources) (Section 3.1). It also corrected one earlier provisional exclusion: Law et al. (2019), previously set aside as a presumed conference abstract from the same San Diego AB-PA01 program as Aslam, Courtwright, et al. (2019), is in fact a full peer-reviewed report of a distinct MDR cystic-fibrosis patient, and is now included. Four records were documented as exclusions: Van Nieuwenhuyse et al. (2022), a single XDR case already counted within Pirnay et al. (2024)’s 100-case Belgian-consortium cohort (the same de-duplication basis as the Dan et al. (2023) exclusion in Section 2.4); Zurabov et al. (2023), an ICU phage-monotherapy case series whose *P. aeruginosa* outcomes cannot be separated from a multi-pathogen cohort (otherwise one of the few monotherapy sources anywhere in this literature); Hayakawa et al. (2025), a multi-pathogen feasibility study with no *P. aeruginosa*-separable outcome; and Chung et al. (2025), a perspective article rather than a primary study. With three databases searched — and because Scopus indexes much of EMBASE’s content — this review meets a recognized minimum for a defensible PRISMA search. EMBASE and Web of Science were nonetheless not searched, and we state plainly why: this review team has no institutional subscription to either, so those searches could not be executed rather than merely not having been completed yet. So that any reader with subscription access can run them and audit what, if anything, they add, we report the adapted queries in full here rather than leaving them undocumented. For Ovid EMBASE:

```
(pseudomonas aeruginosa or P. aeruginosa).ti,ab. AND (phage or bacteriophage or
phage therapy or bacteriophage therapy).ti,ab. AND (multidrug-resistant or MDR
or extensively drug-resistant or XDR or pandrug-resistant or PDR or drug-resistant
or drug resistance).ti,ab. AND (2016:2026).yr.
```

and for Web of Science Core Collection:

```
TS=((("Pseudomonas aeruginosa" OR "P. aeruginosa") AND (phage OR bacteriophage OR
"phage therapy" OR "bacteriophage therapy") AND ("multidrug-resistant" OR MDR OR
```

```
"extensively drug-resistant" OR XDR OR "pandrug-resistant" OR PDR OR "drug-resistant"  
OR "drug resistance")) AND PY=(2016-2026)
```

We take two partial mitigations. First, Scopus and EMBASE overlap substantially in biomedical journal coverage, so the marginal yield of EMBASE over an already-completed native Scopus search is smaller than its absence might suggest. Second, in place of the unavailable Web of Science forward-citation function, we hand-screened the reference lists and citing literature of the three most relevant prior syntheses (El Haddad et al., 2019; Uyttebroek et al., 2022; Liu, Thong, et al., 2025) and ran a targeted screen of individually named landmark cases (Section 3.1), which recovered two eligible studies no keyword search had returned. Neither mitigation is equivalent to the searches themselves, and we do not claim otherwise; the gap is a real and permanent constraint on this review, carried forward as a limitation (Section 4.4) rather than as outstanding work.

Finally, one supplementary, non-PRISMA source contributed candidate studies: an ad hoc, single-reviewer title/abstract screen of 201 PDF records assembled outside the formal database search (via a PubMed/Elicit export), screened against this review’s PICO criteria using a four-criterion rubric (population, intervention, study design, and publication-date thresholds all met) replicated from an earlier 76-document reference screen. This pass returned 12 records meeting all four criteria and 189 exclusions, of which 3 of the 12 duplicated studies already in the formal search and 9 were genuinely new candidates, two of which (Krakhotkin et al., 2025 and an immunological sub-study later confirmed to duplicate an already-catalogued case) required subsequent correction (Section 2.4). Because this screen was conducted by a single reviewer, predates protocol registration, and did not use the dual-independent-reviewer process specified as a Field Convention for this review, we treat its disposition as a scoping input requiring independent re-screening or spot-checking, not as a component of the formal PRISMA screening record, and recommend it be re-run under the registered protocol before this review’s flow diagram is finalized.

A second supplementary channel was added during peer review: a targeted landmark-

case screen, prompted by a referee’s observation that several canonical *P. aeruginosa* phage-therapy cases might not have been captured by the keyword searches above. Each named landmark case was screened against this review’s PICO criteria, and two proved eligible and previously uncaptured. The first is Chan, Turner, et al. (2018), the OMKO1 case: an MDR *P. aeruginosa* chronic aortic Dacron-graft infection treated with local phage OMKO1 plus ceftazidime and resolved with no recurrence over 18 months off antibiotics (distinct from Blasco et al. (2023)’s vascular-graft failure and from the Chan et al. 2025 cystic-fibrosis cohort). The second is Arya et al. (2026), an MDR *P. aeruginosa* periprosthetic joint infection treated with intermittent bacteriophage monotherapy — no antibiotic — over roughly two years, with clinical resolution but without microbiological eradication; it is only the second phage-monotherapy arm anywhere in this corpus (Section 3.2). Both were added to the analytic dataset through this documented channel. Two further named cases were screened and not added: Rubalskii et al. (2020), a multi-pathogen phage series whose *P. aeruginosa* outcomes cannot be separated from the mixed cohort (the same **mixed-not-separable** basis as other exclusions in this review), and Cano et al. (2021), a *Klebsiella pneumoniae* prosthetic-joint case that falls outside this review’s pathogen scope — so the absence of both from the corpus is correct, not an omission. Like the PDF screen above, this landmark-case channel is a documented supplementary source, not a component of the formal database-search record.

2.4 Study Selection

Screening and full-text assessment were conducted by a single reviewer at each stage (this review’s data-extraction team), with a second reviewer’s critique applied sequentially after each round rather than concurrently alongside the first screen. This is a deviation from the dual-independent-screening convention this field expects: no discovery-phase artifact for this project documents the personnel capacity for genuine dual-independent screening and extraction, and this review proceeded on a single-reviewer-plus-sequential-critique basis instead.

Two corrections made during screening and extraction illustrate why full-text verification, not abstract- or supplementary-table-level characterization, was necessary before any study entered the analytic dataset. First, Krakhotkin et al. (2025) was initially catalogued as the closest structural match anywhere in this search to the monotherapy-versus-combination stratification this review set out to estimate: a four-arm, prospective observational comparative study of bacteriophage cocktail alone, bacteriophage plus one antibiotic, bacteriophage plus two antibiotics, and a two-drug antibiotic regimen with no bacteriophage, in women with multidrug-resistant chronic recurrent cystitis. It was also misclassified, in the supplementary risk-of-bias table of the largest existing phage-therapy meta-analysis, as a randomized controlled trial (Liu, Thong, et al., 2025). Its own title, confirmed directly against the published abstract, identifies it as an observational comparative study. Full-text verification, obtained after this misclassification was caught, further confirmed that the study enrolled zero *P. aeruginosa* cases: the causative organisms were *Escherichia coli*, *Klebsiella pneumoniae*, *Proteus mirabilis*, *Staphylococcus epidermidis*, and *Staphylococcus aureus* only. Krakhotkin et al. (2025) was excluded from this review’s evidence base in its entirety on this basis; it appears here only as an example of why this review’s own screening caught an error present in a prior team’s dataset, and should not be mistaken for a retained data source.

Second, an immunological sub-study reporting host T-cell and antibody responses to bacteriophage in a lung-transplant recipient on adjunctive phage therapy for multidrug-resistant *P. aeruginosa* pneumonia (Dan et al., 2023) was confirmed, on full-text comparison, to describe the identical clinical case already reported as Patient 1 in Aslam, Courtwright, et al. (2019): the same bilateral lung-transplant recipient, the same two treatment episodes, the same intravenous-plus-nebulized phage regimen. This study was excluded as a confirmed duplicate, avoiding double-counting a single patient’s outcome twice in the pooled estimates.

A third finding from Round 4’s targeted search (Section 2.3) concerns Leitner et al. (2021), the single cleanest genuine antibiotic-monotherapy comparator arm identified anywhere in this literature: a three-arm randomized trial of intravesical bacteriophage versus

placebo versus open-label systemic antibiotics, in which *P. aeruginosa* is confirmed among the enrolled uropathogens. Despite an exhaustive attempt to obtain a *Pseudomonas*-specific subgroup breakdown, using direct PubMed full-text retrieval, a PMC linkout check, a ClinicalTrials.gov structured-results query, and a search for secondary sources citing the trial in enough detail to reconstruct a per-pathogen breakdown, no such breakdown could be located. This trial’s outcome data are reported only as a six-pathogen aggregate with no route toward isolating the *P. aeruginosa* subgroup from the sources available to this review team. It is retained in the extraction dataset as a documented, access-blocked exclusion – an open question, not a settled negative – since full-text or author-correspondence access, if obtained, could still resolve this study into the monotherapy stratum this review otherwise could not populate (Section 2.7).

2.5 Data Extraction

The unit of extraction is the study-arm, not the study: a study reporting outcomes for more than one regimen (for example, more than one administration route within the same cohort) contributes one row per arm. Extraction was conducted by a single reviewer per study, again without concurrent dual-independent extraction, for the same resourcing reason disclosed in Section 2.4. The extraction schema, fixed before extraction began, records for each study-arm: sample size; pathogen scope (*Pseudomonas*-only, mixed-with-separable-subgroup, or mixed-not-separable, the last excluded per Section 2.2); resistance classification and its source; route of phage administration; treatment modality; the numerator and study-specific definition for clinical success; the numerator for any adverse event; microbiological eradication; mortality; length of hospitalization; resistance emergence during treatment; study design; publication year; journal tier; geographic source; and a documented primary-source citation locator (page, table, or verbatim quote) for every extracted value.

Resistance classification followed a four-step ladder applied in order: (1) if the primary study reports a complete antibiogram sufficient to apply the Magiorakos et al. (2012) criteria

directly (non-susceptibility to at least one agent in at least three antimicrobial categories for MDR, to all but two or fewer categories for XDR, or to every agent tested in every category for PDR), the arm is coded **independently-verified**; (2) if the study labels the population MDR, XDR, or PDR under its own stated criteria without a full panel, the arm is coded **author-reported** and the author’s label is used as given; (3) if only a partial panel is reported, an asymmetric, deliberately conservative rule applies: an MDR call is permitted from partial data, but an upgrade to XDR requires positive confirmation of susceptibility across enough categories to rule out a milder classification, not merely the absence of testing in the untested categories; (4) if no classification information is available at all, the arm is coded **not classifiable** and retained as its own explicit category, not merged into MDR or dropped. This asymmetric rule has a mechanical consequence: it shrinks the XDR stratum relative to a symmetric rule, because any partially tested isolate that might plausibly be XDR defaults to MDR or to not-classifiable absent explicit confirmation. This is a deliberate trade-off between classification conservatism and statistical power in the XDR stratum, and a reader comparing this review’s XDR estimate against another synthesis’s should attribute any difference to this coding choice before attributing it to a clinical difference.

Because the largest existing phage-therapy meta-analysis (Liu, Thong, et al., 2025) makes a case-level extraction dataset publicly available as a supplementary file, this review adopted an explicit, tiered reuse policy, stated up front. Tier 1, using that dataset purely to identify candidate studies for citation-chasing (Section 2.3), was used as described above and is standard PRISMA practice, since no outcome data were imported at that stage. Tier 2 calls for independently cross-validating this review’s own extracted values against Liu et al.’s corresponding entries for every study appearing in both datasets, with a documented primary-source citation required for a sample of “independently extracted” rows to guard against a coder simply copying Liu et al.’s values and mislabeling them as independent. This was adopted as the intended default policy but has not yet been executed as a formal cross-validation exercise on this extraction round: every row in the final analytic dataset is coded

`data_provenance = independently-extracted`, and no row in the final analytic dataset carries a `cross-validated-against-Liu2025` label. The planned side-by-side comparison against Liu et al.’s codings for overlapping studies remains outstanding. Tier 3, direct adoption of Liu et al.’s codings in place of independent extraction, was never invoked for any study in this dataset, consistent with the policy that Tier 3 requires explicit, per-instance approval and is expected to be rare.

The final analytic dataset contains 50 study-arms across 42 studies. One arm (Leitner et al., 2021) is excluded from all pooling, for the access-blocked reason given in Section 2.4, leaving 40 pooling-eligible study-arms across 33 studies. The largest single source, Pirnay et al. (2024), reports 49 *P. aeruginosa* patients across a 100-patient multinational cohort. An initial extraction pass captured only an 11-patient subset given full case-narrative treatment in the source paper’s main text; this was superseded during drafting by a full re-extraction from the paper’s open-access supplementary tables (PMC11153159), which report structured per-patient data for all 100 cases. Of the 49 *P. aeruginosa*-targeted patients, 17 are classified “usual drug resistance” in the source data and do not meet this review’s MDR/XDR/PDR eligibility criterion; they are excluded. The remaining 32 patients are extracted as three resistance-stratified arms (7 XDR, 23 MDR, 2 PDR), replacing the original 11-patient aggregate: all 32 eligible patients are now included, resolving the within-study selection-bias risk the narratively-selected 11-patient subset had raised. Cross-validation against that original subset found exact agreement for 9 of 11 patients; the other 2 (case #30 and #71) are now excluded as usual-drug-resistance, a correction made possible only once per-patient resistance data became available, reflecting new information rather than a flaw in the original extraction. A handful of arms have missing values for individual outcomes while the rest of the row remains intact: most notably, clinical success is not extractable as a binary proportion for two RCTs, whose primary endpoint is a continuous or time-to-event measure instead of a fixed-timepoint success count, and likewise for Chan et al.’s (2025) cystic-fibrosis cohort, whose endpoints are continuous ppFEV₁ and sputum-CFU measures rather than a binary

success count; these arms are excluded from that one outcome’s pooling and retained in the dataset for every other outcome.

2.6 Risk-of-Bias Assessment

The pre-specified plan assigns ROBINS-I (Sterne, Hernán, et al., 2016) to non-randomized comparative studies, Cochrane RoB2 (Sterne, Savović, et al., 2019) to randomized controlled trials, and the JBI critical appraisal tool for case series (Munn et al., 2020) to single-arm case series and cohorts, with the case-report-specific approach of Murad et al. (2018) reserved for the smallest reports (one to three patients), since several JBI case-series items presume a scale of reporting that does not apply at that size. ROBINS-I was not applied to any study-arm in the final analytic dataset: every non-randomized source in this corpus is single-arm (a case report, case series, or cohort without a within-study comparator), so none meets ROBINS-I’s own scope of non-randomized *comparative* studies; all such sources were rated with the JBI or Murad et al. tools instead.

Risk of bias is appraised for all 40 pooling-eligible study-arms by pre-specified rule rather than by hand (`scripts/R/12_risk_of_bias.R`), applying the four Murad et al. (2018) domains — selection, ascertainment, causality and reporting — uniformly to every arm, since the population criterion leaves the corpus entirely observational. Each domain is scored from extraction metadata this review already records, and is scored as unclear wherever that metadata is silent, so the appraisal never reads more favourably than the evidence supports. An earlier version of this section reported the single-patient reports as Low-to-moderate risk; that characterisation rested on a per-domain tabulation that was never retained and is withdrawn (Section 3.3). Two consequences of that withdrawal should be stated rather than left implicit. First, the appraisal now returns *High* risk overall for every one of the 40 pooling-eligible arms, because Murad’s causality domain cannot be satisfied wherever phage was co-administered with antibiotics and overall rating is the maximum across domains; the earlier characterisation was not merely unretained but directionally wrong. Second, this

review’s earlier rounds carried RoB2 ratings for four randomised trials and JBI ratings for several cohorts and series. The population criterion (Section 2.2) has since removed every trial arm from pooling, so those ratings now describe arms that contribute to no pooled estimate and no GRADE cell. They are not reported here, because reporting them would describe a synthesis this review no longer performs.

Two elements of that earlier work do carry forward and are recorded for provenance. Pirnay et al.’s full per-patient supplementary data were read in full for this review’s resistance-stratified re-extraction (Section 3.2), and that reading is what supports the arm-level resistance classification of the single largest contributing cohort. The planned reuse of Liu, Thong, et al. (2025)’s existing risk-of-bias assessments was to be gated on independently re-rating at minimum a 20% random sample of overlapping studies, or all studies falling within this review’s pre-specified strata, whichever is larger. That spot-check has not been performed, no study-arm carries an `adopted-from-Liu2025-spot-checked` disposition, and accordingly none of Liu, Thong, et al.’s ratings is used anywhere in this review.

GRADE certainty of evidence (Guyatt et al., 2011) was assessed for every pooled outcome-stratum cell. Because every cell is an uncontrolled single-arm proportion drawn predominantly from compassionate-use case reports and case series — the randomized trials contribute only their phage arms, stripped of their comparators — each body of evidence starts at *Low* certainty, the GRADE default for observational evidence, and is then rated down across the five standard domains of risk of bias, inconsistency, indirectness, imprecision and publication bias (Guyatt et al., 2011). To keep the assessment reproducible rather than a set of unaudited per-cell judgments, the domains are applied by pre-specified rules to the cell’s own data (`scripts/R/10_grade.R`): risk of bias, rated down two levels where every arm contributing to the cell rates High risk of bias overall on Murad et al. (2018) and one level otherwise. Both branches are reached: 38 of the 40 arms rate High, and the 2 phage-monotheapy arms that rate Moderate pull the cells containing them to -1 (Section 3.8). In earlier rounds, when every arm rated High, this domain was a constant and is reported as

such in those versions; inconsistency, rated down two levels where the 95% prediction interval spans at least 80 percentage points and one level at 50 or more (the prediction interval rather than I^2 , which is degenerate for a binomial-normal GLMM and is documented as frequently uninformative and misinterpreted in proportion syntheses; Migliavaca et al., 2022); indirectness, rated down one level for the compassionate-use salvage population common to every cell and two for clinical success, whose definition varies by study across non-comparable syndromes; imprecision, rated down two levels where the 95% confidence interval spans at least 50 percentage points and one at 30 or more; and publication bias, rated down one level in every cell, since a compassionate-use case report describing a failure is markedly less likely to be written and published than one describing a success, and Peters’ test cannot exclude this at the present corpus size.

Three of these five domains — risk of bias, indirectness, and publication bias — deduct at least one level in every cell by construction, so the minimum possible deduction is three levels against a starting certainty of Low. *Every cell therefore rates Very low before the data-driven domains are consulted*, and we report the rating as deterministic rather than presenting a uniform floor as though it were a discriminating finding. Only inconsistency and imprecision vary across cells; the resulting domain profile, not the rating, is what distinguishes one cell from another (Section 3.8). No cell qualified for the observational upgrade criteria (large magnitude, dose-response, or confounding that would work against the observed direction), none of which is estimable without a comparator. Certainty cannot fall below *Very low*, GRADE’s floor.

2.7 Outcomes and Statistical Analysis

Four outcome families are pooled where the evidence permits: clinical success and any adverse event (primary), and microbiological eradication and mortality (secondary); length of hospitalization and resistance emergence during treatment were extracted but are reported narratively rather than pooled, given their sparse and heterogeneous reporting across studies.

Each outcome is pooled separately within every cell of a pre-specified three-way stratification: resistance category (MDR, XDR, PDR, or not classifiable), route of phage administration (topical/local, intravenous, inhaled/nebulized, oral, intra-articular, or other), and treatment modality (phage monotherapy, phage-antibiotic combination, or antibiotic monotherapy). An unstratified, pooled-across-everything estimate is also reported for each primary outcome, explicitly labeled secondary and contextual, for comparability with prior non-stratified syntheses (El Haddad et al., 2019; Uyttebroek et al., 2022; Liu, Thong, et al., 2025). This review’s contribution is the stratified breakdown, not a single headline proportion.

The primary pooling model is a random-effects binomial-normal generalized linear mixed model (GLMM) with a logit link (Hamza, Houwelingen, and Stijnen, 2008; Nyaga, Arbyn, and Aerts, 2014), implemented in R (version 4.6.1) using the `meta` and `metafor` packages:

$$r_i \mid p_i \sim \text{Binomial}(n_i, p_i), \quad \text{logit}(p_i) = \mu_s + u_i, \quad u_i \sim N(0, \tau_s^2),$$

with the pooled proportion for stratum s recovered as $\hat{p}_s = \text{expit}(\hat{\mu}_s)$ and its 95% confidence interval computed on the logit scale before back-transformation. We chose this model over the Freeman-Tukey double-arcsine transformation specified as the field’s default variance-stabilizing approach for proportions near the 0/1 boundary (Freeman and Tukey, 1950; Miller, 1978), because this corpus spans study-arms as small as a single case report and as large as a 100-patient cohort, and Schwarzer et al. (2019) document that double-arcsine back-transformation under exactly this kind of sample-size heterogeneity can produce a pooled point estimate lying outside the range of every observed study. The GLMM avoids this failure mode by modeling the exact binomial likelihood at each study-arm’s own sample size, with no transformation or back-transformation step. Double-arcsine (DerSimonian-Laird τ^2) and a simple logit transformation with DerSimonian-Laird pooling (DerSimonian and Laird, 1986) are reported alongside the primary model as sensitivity comparisons, and the Hartung-Knapp-Sidik-Jonkman adjustment (IntHout, Ioannidis, and Borm, 2014) is requested for

every stratum’s confidence interval, given the small number of studies expected per cell (Barker et al., 2021). We distinguish requesting the adjustment from its binding, because in this corpus it mostly does not bind. The Hartung-Knapp estimator multiplies the standard error by a factor that reduces to one when the between-study variance estimate sits at zero. A diagnostic computed for every pooled cell (`scripts/R/05_robustness.R`) compares the model’s reported standard error against the naive complete-pooling binomial standard error on the logit scale, $\sqrt{1/r + 1/(N - r)}$: the ratio is exactly one in 11 of the 13 pooled cells, and departs from one only in the two cells where $\tau^2 > 0$, where it reaches 2.92. What the manuscript reports for those 11 cells is therefore a t -interval on the logit of the pooled counts, not a variance-inflated random-effects interval, and it is described as such in Section 3.6 rather than under a label the arithmetic does not support.

Because a binomial-normal GLMM fit by numerical quadrature can fail to converge in exactly the sparse, boundary-proportion regime this corpus produces (many single-patient case reports with zero or one event, and strata where every arm reports 0% or 100% of an outcome), every stratum-outcome cell’s GLMM fit was checked for convergence (captured errors, known non-convergence warning strings, or a non-finite pooled estimate or standard error), and this diagnostic is reported for every cell, not only the cells that failed. Where GLMM failed to converge for a specific cell, the logit-DerSimonian-Laird sensitivity model was promoted to serve as that cell’s reported primary estimate; this substitution is cell-level only, never model-wide, and is always disclosed with an explicit footnote naming the fallback and the reason, so that a fallback estimate is never visually indistinguishable from a converged GLMM estimate. Where a single study contributed more than one arm to the same stratum-outcome cell, robust variance estimation clustered by study (`clubSandwich`) was applied alongside the naive-independence estimate, since arms from the same study share a site, a measurement protocol, and often the same investigators. A stratum-outcome cell is reported as a formal pooled estimate only if it contains at least 3 independent studies and at least 20 pooled patients; cells below this pre-specified threshold are reported narratively instead:

individual study-level proportions are tabulated with an explicit statement that the evidence is insufficient to support formal pooling, so an unstable estimate from one or two studies is never presented as though it were a stable finding. Between-stratum comparisons use a subgroup meta-analysis with Cochran’s Q -between test on the transformed proportion scale, not a meta-regression on a paired treatment effect, since no paired effect size exists under this design; no formal correction for multiple comparisons (Bonferroni, Benjamini-Hochberg) is applied across the resulting battery of subgroup tests, a deliberate choice consistent with the Cochrane Handbook’s treatment of pre-specified subgroup analyses as hypothesis-generating rather than confirmatory (Higgins et al., 2023), since the stratification variables were fixed for clinical reasons before extraction, not chosen after seeing which splits looked significant.

Where a genuine within-study phage-versus-antibiotic-monotherapy comparator exists, a paired risk ratio was planned as a clearly labeled exploratory supplementary analysis, never blended into the primary stratified-proportions estimand. In the event, no such comparison is reported: Krakhotkin et al. (2025) is excluded entirely (Section 2.4), Leitner et al. (2021) is excluded from pooling for lack of a *Pseudomonas*-specific breakdown, and the one remaining candidate (PhagoBurn) reports a time-to-event, not a binary, primary outcome. This exploratory analysis was planned but ultimately not reportable given the data this review could extract.

Robustness was assessed through thirteen pre-specified checks: transformation and model choice (GLMM versus double-arcsine versus logit-DerSimonian-Laird versus fixed-effect); restricting resistance classification to independently verified cases only; fixed- versus random-effects pooling; a study-design-stratified comparison (randomized and prospective designs versus retrospective and compassionate-use designs); a geographic-concentration check excluding the two most accessible cohorts (Belgium, Israel) to test whether the pooled estimate is an artifact of which data happened to be easiest to obtain; leave-one-out removal of each study; a collapsed two-category route taxonomy (systemic versus local) alongside the full route taxonomy; a transparency check comparing the pooled estimate using only inde-

pendently extracted rows against all rows, given the Liu et al. reuse policy in Section 2.5; Hartung-Knapp-Sidik-Jonkman versus standard Wald confidence intervals; robust-variance clustering for same-study multi-arm cells; a funnel plot and Peters’ regression test for small-study effects—Peters’ test, regressing the event proportion on the inverse sample size, rather than Egger’s linear regression, because for a meta-analysis of proportions the standard error is a deterministic function of the proportion itself and manufactures funnel asymmetry near the 0/1 boundary, so Egger’s test is invalid here and Peters’ is the recommended small-study test for binary/proportion outcomes (Peters et al., 2006). We flag the provenance of this choice rather than let it pass as pre-specified: the pre-analysis plan named Egger’s test, Egger’s was run, and Peters’ replaced it *after* that result was seen (Section 3.7). It is therefore a **post-hoc analytic change**, adopted because Egger’s is invalid for proportions, not because it was planned. Readers should weigh it accordingly. Peters’ also has little power in this corpus by construction: its regressor is the inverse sample size, and because 33 of the 40 pooling-eligible arms contain a single patient, that regressor takes the identical value 1 for roughly two-thirds of the sample, leaving the regression to be driven by a handful of larger arms—applied only to stratum-outcome cells with at least 10 studies per the pre-specified threshold, and explicitly not run, with that reason stated, below it; an appendix-only relaxation of the minimum-pooling threshold to at least 2 studies and 10 patients, to show whether the primary threshold is itself consequential without promoting the relaxed-threshold estimates to primary results; and application of the pre-specified monotherapy-stratum decision rule once full-text access to the candidate comparator studies was exhausted.

Six falsification checks, adapted from causal-inference negative-control logic to this descriptive design, were pre-specified to test whether the pooled proportions reflect a genuine, interpretable summary of the evidence rather than an artifact of reporting drift or selective inclusion: a meta-regression of the transformed clinical-success proportion on publication year, expecting a null or weak coefficient absent a documented mechanism; a comparison of pooled proportions across journal tiers, since a strong tier gradient would itself suggest editorial or

selective-reporting distortion rather than a clinical finding; a visual small-study/sample-size check plotting each arm’s proportion against $1/\sqrt{n}$, alongside the formal Peters’ regression test; a cross-tabulation of route by resistance class by geographic source, to rule out an apparent route effect that is actually a single-center signature; a comparison of length-of-hospitalization against clinical-success and eradication, on the logic that if every outcome, including one with no obvious direct phage mechanism, moves favorably to a similar degree, that pattern is more consistent with a uniform reporting or selection artifact than with a phage-specific signal; and a check of adverse-event rates against publication year, to rule out apparent safety improvement that is actually a change in adverse-event ascertainment practice over time. The length-of-hospitalization comparison is reported as not applicable in the current analysis: only 1 of the 31 pooling-eligible study-arms reports a length-of-hospitalization value, too sparse to support even a qualitative directional comparison against the clinical-success or eradication estimates.

3 Results

3.1 Study Selection

The primary electronic search (Section 2.3) returned 54 raw PubMed/MEDLINE hits on the original comparative-design question, 48 further hits from the systematic-review/meta-analysis recheck, and 10 ClinicalTrials.gov records, screened individually across four rounds (2026-07-18 to 2026-07-20). Round 3’s citation-chasing pass against the largest existing phage-therapy meta-analysis’s risk-of-bias table (Liu, Thong, et al., 2025) surfaced five uncatalogued records — Krakhotkin et al. (2025), Leitner et al. (2021), Karn et al. (2024), Rhoads et al. (2009), and Stanley et al. (2025) — of which only Leitner et al. entered the analytic dataset (extracted, excluded from pooling; Section 2.4); Krakhotkin et al. was excluded outright (zero *P. aeruginosa* cases); Rhoads et al. failed eligibility on a structural ground (saline, not antibiotic, comparator); Karn et al. and Stanley et al. were both resolved

at peer review: Karn et al. is a chronic-wound trial reporting only on multidrug-resistant bacteria generically, with no *Pseudomonas* breakdown and no *Pseudomonas* MeSH term, and Stanley et al. is the Yale CYPHY trial (NCT04684641), excluded under the population rule because its inclusion criteria required only a positive culture. Round 4’s targeted antibiotic-only-cohort search returned 44 hits, none containing a phage arm — confirming, not resolving, the absent monotherapy comparator. A supplementary, non-PRISMA, single-reviewer screen of a 201-document PDF corpus contributed 9 further candidates after excluding 3 duplicates; 2 of the 9 required correction on full-text review, Krakhotkin et al. (above) and an immunological sub-study (Dan et al., 2023) confirmed to duplicate Patient 1 in Aslam, Courtwright, et al. (2019) – net 8 unique new records once Krakhotkin’s cross-channel duplication with the citation-chasing pass is resolved. The remaining six included studies (Pirnay et al. (2024), Onallah, Hazan, Nir-Paz, et al. (2023), Weiner et al. (2025), Jault et al. (2019), the Phase 1b/2a inhaled-phage cystic-fibrosis trial (*SWARM-P.a. (AP-PA02): Phase 1b/2a Randomized, Double-Blind, Placebo-Controlled Trial of Inhaled Phage Therapy in Cystic Fibrosis Pseudomonas aeruginosa* 2022), and Aslam, Courtwright, et al. (2019)) were identified during this review’s initial discovery-phase literature scoping, an earlier WebSearch/WebFetch-based pass that predates the four documented, structured-query rounds above and was not itself run with recorded, countable hits per query – a disclosed methodological gap, not a hidden one (Figure 1). A broad topical PubMed/MEDLINE search (481 records over the pre-specified 2016–2026 window; Section 2.3), run to place PubMed on the same topical footing as Scopus rather than only the narrow comparative-design rounds above, recovered the eligible clinical studies already identified but added no unique-new record; a dedicated screen of its 2016–2018 sub-window found only preclinical work beyond the already-included Duplessis et al. (2018) and PhagoBurn. Finally, a native Scopus database search (860 records screened; 2026-07-23; Section 2.3) — the third core database alongside PubMed/MEDLINE and ClinicalTrials.gov — substantially enlarged the corpus, adding ten eligible studies the earlier rounds had missed: Köhler et al. (2023), a single

inhaled-phage MDR case; Chan et al. (2025), a nine-patient nebulized-phage cystic-fibrosis cohort split into an MDR (n=7) and a PDR (n=2) arm; and eight further human clinical *P. aeruginosa* phage-therapy reports (Law et al. 2019; Hahn et al. 2023; Levêque et al. 2023; Duplessis et al. 2018; Denis et al. 2026; Malhotra et al. 2026; Yang et al. 2025; Li et al. 2025 — an 88-year-old woman with an MDR biliary-tract infection, previously carried as a pending record and now resolved from two concordant secondary sources). The native search also corrected one earlier provisional exclusion — Law et al. (2019), previously set aside as a presumed conference abstract from the same San Diego AB-PA01 program as Aslam, Courtwright, et al. (2019), is a full peer-reviewed report of a distinct MDR cystic-fibrosis patient — and documented four exclusions: Van Nieuwenhuyse et al. (2022), a single XDR case already counted within Pirnay et al. (2024)’s Belgian-consortium cohort; Zurabov et al. (2023), an ICU phage-monotherapy case series whose *P. aeruginosa* outcomes cannot be separated from a multi-pathogen cohort (otherwise a rare monotherapy source); Hayakawa et al. (2025), a multi-pathogen feasibility study with no separable *P. aeruginosa* outcome; and Chung et al. (2025), a perspective article rather than a primary study. That a native Scopus search roughly doubled the pooling-eligible corpus (from 13 to 23 studies) confirms the earlier PubMed-only coverage gap was real and material, and validates searching Scopus natively rather than through a federated adjunct. Finally, a peer-review landmark-case screen (Section 2.3) added two further eligible studies that no keyword search had captured — Chan, Turner, et al. (2018), the canonical OMKO1 case of an MDR aortic Dacron-graft infection (local phage plus ceftazidime, resolved without recurrence), and Arya et al. (2026), an MDR periprosthetic joint infection treated with phage monotherapy — bringing the pooling-eligible corpus to 25 studies at that point, while confirming that Rubalskii et al. (2020, a multi-pathogen series with no *Pseudomonas*-separable outcome) and Cano et al. (2021, a *Klebsiella* case) are correctly outside it. A fifth screening pass followed directly from peer review. Both round-5 referees identified the search string, not the databases, as the binding constraint: every executed query required a resistance term in title or abstract, while

this review’s Population criterion explicitly *retains* arms with no resistance documentation at all (Section 2.2). A search cannot require a keyword to find studies it then includes on the ground that the keyword was never recorded. Screening the four studies one referee named confirmed the criticism. Zaldastanishvili et al. (2021) contributes two eligible arms and Green et al. (2023) one, and none of the three records carries a resistance term in its title or abstract, so no query this review had run could have retrieved them. Onsea et al. (2019) was excluded at full text: its one XDR *P. aeruginosa* arm was *non-susceptible* to the anti-Pseudomonas phages in the administered cocktail, so it measures nothing about phage therapy against this review’s target organism, and it is additionally a Queen Astrid Military Hospital case with Pirnay et al. as a co-author — the profile this review de-duplicates against. The fourth, a case attributed to Doub, does not exist as a distinct clinical report: the Doub-authored *P. aeruginosa* case indexed in PubMed is Arya et al. (2026), already in this corpus. Two eligible studies came out of that batch — Zaldastanishvili et al. (2021) with two arms and Green et al. (2023) with one — and all three arms are *eradication failures*.

That result made the referee’s second demand unavoidable, and it carried a falsifiable bar: re-run the search with the resistance block removed, and if it yields two or fewer new eligible studies, accept that the corpus is near-complete. We ran it. PubMed as title/abstract *Pseudomonas aeruginosa* with phage-therapy terms, restricted to clinical publication types and 2016–2026, returned 45 records: 17 already traceable to this corpus and 28 never screened. The bar was not met, and by a wide margin: nine of the 28 entered the analytic dataset and six more were assessed at full text and excluded. The disposition of each is stated rather than summarised.

Four are pooled. Casazza et al. (2025), chronic mastoiditis from an MDR strain in a lung-transplant recipient, is the only one that achieved eradication. Teney et al. (2024), four successive episodes of ventilator-associated pneumonia in a patient with burns over 81% of the body surface, did not: its conclusion states verbatim that phage-antibiotic synergy “did not result in microbial success” while the clinical outcome was favourable. Li et al. (2023),

the first-in-human use of a double-stranded RNA phage, did not either — the paper carries a passage headed “reasons for the failure to completely eradicate” — and it contributes only the second phage-attributable adverse event in this review, a fever to 38.7°C after every course. Chen et al. (2022), a post-pneumonectomy empyema, cleared its pathogen.

Two further arms entered after the batch above. Pardo-Freire et al. (2025), a lung-transplant recipient with cystic fibrosis in acute rejection, did not eradicate: nebulised phage produced a 1.2 log reduction and the bacterial load returned to pre-treatment levels by day 25, the later culture-negativity following three weeks of intravenous ceftazidime-avibactam rather than the phage. And Rubalskii et al. (2020) re-enters, which is a correction rather than an addition. This review had excluded it as multi-pathogen and not separable; the domain referee challenged that call and asked which table had been consulted. Table 1 does disaggregate by patient, and its Patient 8 — a monomicrobial *P. aeruginosa* thoracotomy-wound infection after double lung transplantation, treated with phage embedded in fibrin glue and reported individually as healed and culture-negative — was separable all along. Patient 1 is polymicrobial and remains out. That is the second referee challenge to an exclusion upheld in this round, and both are recorded here rather than absorbed silently.

Three were extracted and then excluded, each by a rule already in force, and naming them matters more than counting them. Chung et al. (2026) and Ronit et al. (2024) fail the population criterion on their own published antibiograms: the first isolate was merely intermediate to ciprofloxacin and susceptible to piperacillin-tazobactam, ceftazidime-avibactam and meropenem; the second was resistant to two agents and susceptible to meropenem and ciprofloxacin. Neither reaches Magiorakos et al. (2012)’s three-category threshold, and neither paper claims it does — the same ground on which Liu, Lu, et al.’s first patient was excluded. Jennes et al. (2017) is the harder case and the more instructive one: it is the study the referee named as unfindable by any query requiring a resistance keyword, since its title says “colistin-only-sensitive” rather than XDR, and it would have been this review’s fourth phage-monotherapy arm and a death. It is excluded as a probable duplicate of a

patient inside the Pirnay et al. roster, on the same rule applied to Ferry, Kolenda, Laurent, et al. and Racenis et al., and is restored in the de-duplication sensitivity analysis.

The honest summary of both passes is not that the corpus is now complete. It is that a search this review had called “materially improved” surfaced seven previously unscreened eligible or extractable records as soon as one keyword was dropped, that the referees’ diagnosis of the cause was exactly right, and that the resistance-blocked queries reported in Section 2.3 should be read as a lower bound on what the published record contains. A second peer-review screen assessed three further human clinical *P. aeruginosa* phage reports named as potentially missing, with three different outcomes. Ferry, Kolenda, Batailler, et al. (2021) — an 88-year-old man with a relapsing prosthetic knee infection treated by arthroscopic debridement with local phage injection (the PhagoDAIR procedure), resolved and walking without pain at one year — is eligible and is *added* to the corpus; we confirmed it is not a duplicate of Ferry, Kolenda, Laurent, et al. (2022) — which is in the analytic dataset but, as set out below, is itself excluded from pooling as a probable consortium duplicate — and which reports a pandrug-resistant *spinal* infection treated intra-discally and intravenously over 21 months, a different site, procedure, antibiotic regimen and follow-up. Maddocks et al. (2019) is eligible and is likewise *added*, by a route worth documenting. Its primary text is paywalled, carries no abstract in PubMed and has no PMC deposit, and four retrieval attempts failed, so it was initially recorded under PRISMA’s *reports not retrieved* category. Its outcome data were then recovered from two independent secondary syntheses that tabulate it — Vaezi et al. (2024) and Mitropoulou et al. (2022) — applying the same two-concordant-source standard this review used for Li et al. (2025). The two agree on phage product, duration, both administration routes, all three concomitant antibiotics, and outcome: a 77-year-old woman with ventilator-associated pneumonia and empyema complicated by a bronchopleural fistula after thoracotomy, treated with the four-phage AB-PA01 cocktail intravenously and by nebulizer every 12 hours for seven days alongside gentamicin, ciprofloxacin and ceftolozane–tazobactam, whose oxygenation improved within three days

with successful ventilator weaning and whose cultures were negative at six-month follow-up. Resistance is coded not classifiable, and the reasoning needs stating precisely because it sits close to a rule that would instead require exclusion. Vaezi et al.’s table labels five other rows explicitly “MDR” and records this one as plain *P. aeruginosa*, which is suggestive; but a secondary source declining to apply a label is *not* the same thing as reported susceptibility data failing the Magiorakos et al. (2012) threshold, and only the latter triggers exclusion under the first rule in Section 2.2. No susceptibility panel for this patient is available to us at all. The arm therefore falls under the third rule — a salvage report whose panel is unreported — and is retained rather than excluded. We flag the closeness of the call: a reader who regards a specialist review’s refusal to apply the MDR label as evidence about the isolate, rather than as evidence about the reporting, would exclude this arm, and the population-eligibility sensitivity analysis (Section 3.6) bounds what that choice would cost. A third record, reported to this review as a 2020 single-centre series of consecutive intra-venous phage cases, could not be located in PubMed across four differently phrased queries and is therefore not screened; we state this rather than assert an exclusion we did not perform. The clear implication, which we accept, is that referee-named candidates keep yielding eligible studies, and the search remains under-sensitive (Section 4.4).

The final analytic dataset contains 38 study-arms across 31 studies, of which seven are excluded from all pooling on four distinct grounds. Leitner et al. (2021) is a six-pathogen, trial-wide aggregate with no *Pseudomonas*-specific breakdown (Section 2.4). Patient 1 of Liu, Lu, et al.’s two-patient perinephric-abscess series fails the population criterion on reported data: its antibiogram records non-susceptibility to imipenem alone — one agent of roughly eleven tested — short of the three-antimicrobial-category threshold defining MDR (Magiorakos et al., 2012), notwithstanding the report’s own “antibiotic-resistant” framing; the series’ second patient (PDR) is retained. The remaining three exclusions are the review’s three randomized trials, removed together under the population rule set out in Section 2.2: PhagoBurn (Jault et al., 2019) enrolled burn-wound patients with no MDR requirement, and

the BX004-A (Weiner et al., 2025) and SWARM-P.a./AP-PA02 (*SWARM-P.a. (AP-PA02): Phase 1b/2a Randomized, Double-Blind, Placebo-Controlled Trial of Inhaled Phage Therapy in Cystic Fibrosis Pseudomonas aeruginosa* 2022) trials enrolled cystic-fibrosis patients on chronic pulmonary colonization. None selected for resistance, so none was estimating this review’s estimand; all three are retained as external comparative evidence in Section 4.2 rather than as single-arm proportions here. Two further arms are excluded on a fourth ground, duplication against the Belgian consortium cohort (Section 2.2): Ferry, Kolenda, Laurent, et al. (2022) and Racenis et al.’s left-ventricular-assist-device case each match a case inside Pirnay et al.’s 100-consecutive-case roster closely enough that duplication cannot be excluded, so the cohort entry is retained in preference to the case report. That crosswalk could not use country or treatment year, because the roster publishes neither; it is a judgement on clinical fields rather than a determination, and Section 3.6 reports what the estimates become if both arms are restored. This leaves **40 pooling-eligible study-arms across 33 studies**, contributing 91 patients — a corpus that is now entirely compassionate-use and salvage reporting, and in which resistance is positively established for 28 of the 40 arms, leaving 12 arms and 26 patients (29% of the corpus) with no resistance information at all. Figure 1 sets out the selection funnel. Its terminal counts reconcile exactly against the cleaned extraction dataset and the exclusion log in `02_data_preparation.R`: 50 analytic arms across 42 studies, seven arms excluded on the four grounds above, leaving 40 arms across 33 studies. Its identification-stage counts do *not* reconcile arithmetically to that total, and we state why rather than presenting a figure that appears to balance: this review’s earliest channel was a discovery-phase scoping pass run before the structured queries, without recorded per-query hit counts (Section 2.3), so the path from records-screened to studies-included is auditable for every later channel but not for that one. (Pirnay et al.’s cohort contributes 3 of these 40 arms, not 1 – see Section 3.2 for the resistance-stratified re-extraction that produced this split.)

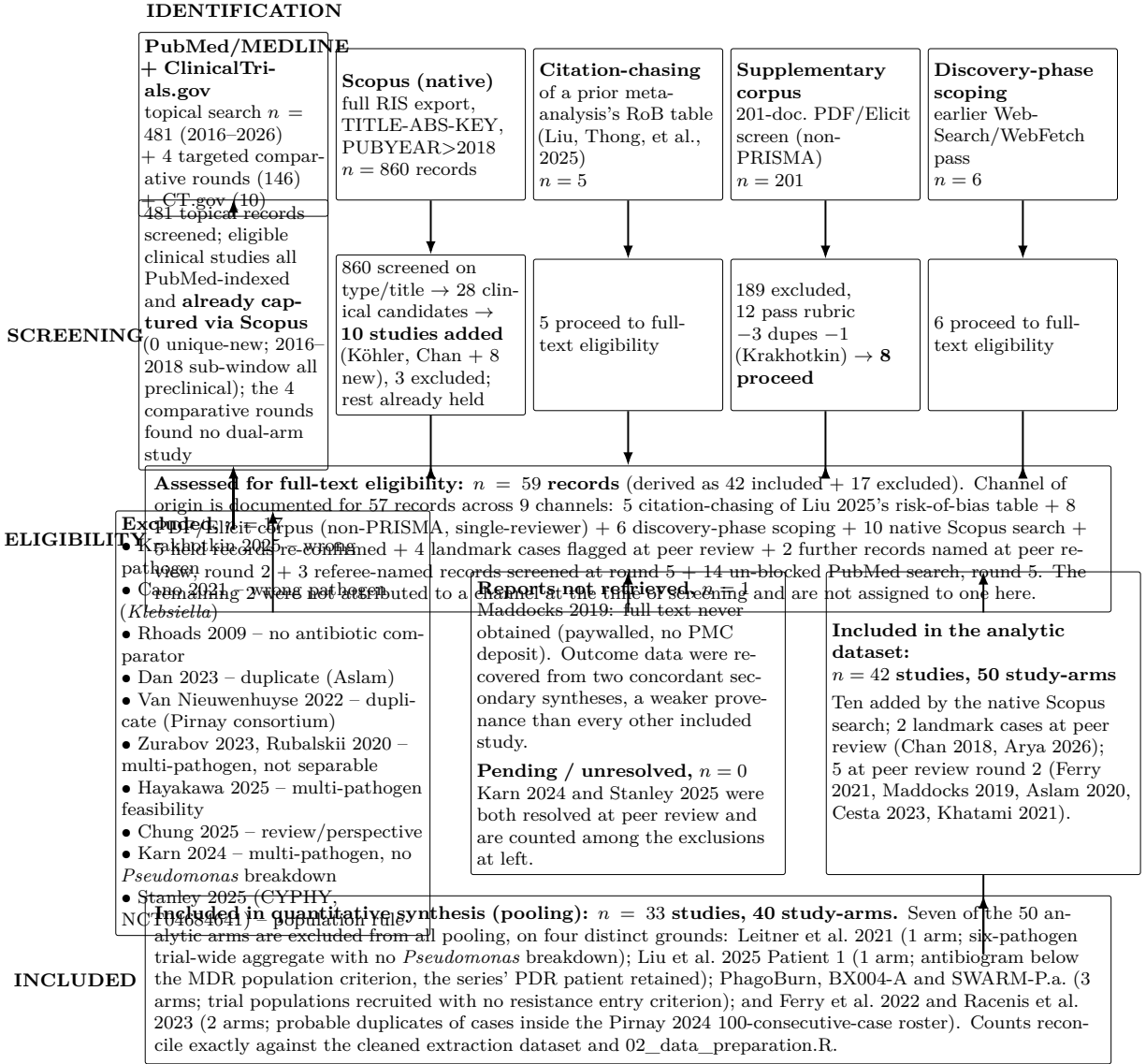


Figure 1: PRISMA 2020-adapted flow diagram of study identification and selection. This review's identification process departed from a single linear database search: multiple channels contributed candidate records – a structured, documented PubMed/MEDLINE and ClinicalTrials.gov search across four rounds (Section 2.3), a native Scopus database search (860 records, 2026-07-23) that added ten eligible studies (Köhler et al. 2023; Chan, Stanley, et al. 2025; and Law et al. 2019, Hahn et al. 2023, Levêque et al. 2023, Duplessis et al. 2018, Denis et al. 2026, Malhotra et al. 2026, Yang et al. 2025, and Li et al. 2025), corrected the earlier provisional exclusion of Law et al. (2019), and documented exclusions (Van Nieuwenhuijse et al., 2022; Zurabov et al., 2023; Hayakawa et al., 2025; Chung et al., 2025), citation-chasing of a prior meta-analysis's risk-of-bias table (Liu, Thong, et al., 2025), a supplementary single-reviewer screen of a 201-document PDF/Elicit corpus (a non-PRISMA source, explicitly flagged as such in Section 2.3), this review's earlier, less formally documented discovery-phase literature scoping, and a peer-review landmark-case screen that added two further eligible studies (Chan et al., 2018 [OMKO1]; Arya et al., 2026) and confirmed two more exclusions (Rubalskii et al., 2020, multi-pathogen; Cano et al., 2021, *Klebsiella*). One record (Krakhotkin et al., 2025) was independently surfaced by two channels and is counted once, at the point its cross-channel duplication was recognized during full-text eligibility review. Two records (Karn et al., 2024; Stanley et al., 2025) were resolved at peer review and are counted among the exclusions: Karn as multi-pathogen with no *Pseudomonas* breakdown, and Stanley (CYPHY) once identified as NCT04684641 and failing the population rule. An earlier version of this caption described both as outstanding while the figure described both as excluded, and counted them in neither box. All counts are traceable to Sections 2.3 and 3.1 and to

3.2 Study Characteristics

The 40 pooling-eligible arms span publication years 2018–2026 and multiple countries or multicentre consortia. Design is now entirely observational and predominantly single-patient: 19 case-report studies contributing 20 arms (Tkhilaishvili et al. (2020), Blasco et al. (2023), Ferry, Kolenda, Laurent, et al. (2022), Ngaay et al. (2026), Racenis et al. (2022), Racenis et al. (2023), Aslam, Courtwright, et al.’s 2019 two lung-transplant patients, Köhler et al.’s (2023) single inhaled-phage case, the seven single-patient reports added by the native Scopus search — Law et al. 2019, Levêque et al. 2023, Duplessis et al. 2018, Denis et al. 2026, Malhotra et al. 2026, Yang et al. 2025, and Li et al. 2025 — Khatami et al. (2021)’s seven-year-old with chronic osteoarticular infection, and the two single-patient landmark cases added through the peer-review landmark-case screen, Chan, Turner, et al. (2018) and Arya et al. (2026)), 5 case series contributing 8 arms (the retained PDR patient from Liu, Lu, et al.’s 2025 perinephric-abscess series, Onallah, Hazan, Nir-Paz, et al.’s 2023 PASA16 series, Chan et al.’s (2025) nine-patient cystic-fibrosis cohort split into an MDR and a PDR arm, and Hahn et al.’s (2023) two-patient cystic-fibrosis series), and 1 retrospective cohort contributing 3 resistance-stratified arms (Pirnay et al. 2024). Arm size ranges from single patients (33 of 40 arms) to 23 (Pirnay’s MDR-stratified arm), 15 (the Onallah PASA16 series), and the 9-patient Chan cystic-fibrosis cohort (split 7 MDR + 2 PDR); the 7 arms with more than one patient together account for 58 of the corpus’s 91 patients.

Pirnay et al.’s cohort was re-extracted in full for this review (Section 2.5): the paper’s open-access supplementary tables (PMC11153159) report per-patient outcomes for all 100 consecutive cases, of which 49 targeted *P. aeruginosa* specifically – matching the paper’s own stated figure exactly. Of those 49, 17 are classified “usual drug resistance” in the source data and therefore do *not* meet this review’s MDR/XDR/PDR eligibility criterion; they are excluded. The remaining 32 patients split into three resistance-stratified arms: 7 XDR, 23 MDR, and 2 PDR. This supersedes an earlier 11-patient subset drawn only from the paper’s main-text case narratives; cross-validation against that subset found exact agreement on

9 of 11 patients; the other 2 (case #30 and #71) are usual-drug-resistance rather than MDR/XDR/PDR and are now excluded. This is a genuine eligibility correction, made possible only once per-patient resistance data became available – new information, not an error in the original 11-patient read.

Resistance classification (Section 2.5) assigns 20 arms MDR, 5 XDR, 3 PDR, and 12 not-classifiable; 3 of the 20 MDR calls and 2 of the 5 XDR calls were independently re-derived from a reported antibiogram, the remainder resting on the primary study’s own label (including all three of Pirnay’s new resistance-stratified arms, which rest on that paper’s own per-patient antibiogram-based classification, not an independent re-derivation by this review team – see the classification-sensitivity check, Section 3.6). Route (collapsed) splits 14 other (multi-route), 7 inhaled/nebulized, 8 intravenous, 7 topical/local, and 4 of unspecified route (Malhotra et al. 2026 and Arya et al. 2026, route not reported in the available source). Modality is overwhelmingly combination therapy (37 of 40 arms). 3 arms are phage monotherapy: Arya et al. (2026), an intermittent phage-only regimen for a periprosthetic joint infection, and the two Eliava Phage Therapy Center patients in Zaldastanishvili et al. (2021), both of whom came to that centre expressly to replace antibiotics with phages and did — one taking antibiotics once in three years during an intercurrent viral illness, the other refusing them outright. No arm is coded pure antibiotic monotherapy. Table 1 tabulates all 50 study-arms across 42 studies at the arm level, including the seven arms excluded from pooling.

3.3 Risk of Bias

Risk of bias is appraised for every one of the 40 pooling-eligible arms, and the appraisal is generated by rule from recorded extraction metadata (`scripts/R/12_risk_of_bias.R`) rather than asserted, so it regenerates with the data and can be audited line by line (Table 2). Because the population criterion removed every randomised trial from the synthesis, a single instrument now applies throughout: the four domains of Murad et al. (2018) for case reports and series — selection, ascertainment, causality and reporting. We state at the outset what

Table 1: Study characteristics, all 50 extracted study-arms

Study	Country/region	Design	<i>n</i>	Resistance	Route	Modality	Pooling
Tkhilaishvili 2020	Germany	Case rpt.	1	XDR	Topical	Combo	Pooled
Blasco 2023	Spain	Case rpt.	1	MDR	IV	Combo	Pooled
Ngauy 2026	USA	Case rpt.	1	XDR	IV	Combo	Pooled
Ferry 2022	France	Case rpt.	1	PDR	Other	Combo	Excluded ^a
Liu 2025	China	Case series	1	NC	Topical	Combo	Excluded ^a
Liu 2025	China	Case series	1	PDR	Topical	Combo	Pooled
Racenis 2023	Latvia	Case rpt.	1	MDR	Other	Combo	Excluded ^a
Racenis 2022	Latvia	Case rpt.	1	MDR	Topical	Combo	Pooled
Onallah 2023	Israel	Case series	15	NC	Other	Combo	Pooled
Weiner 2025	Multi.	RCT	7	NC	Inhaled	Combo	Excluded ^a
Jault 2019	Multi. (EU)	RCT	13	NC	Topical	Phage mono	Excluded ^a
Leitner 2021	Georgia	RCT	28	NC	Other	Phage mono	Excluded ^a
SWARM-P.a. 2022	Multi. (CF)	RCT	10	NC	Inhaled	Combo	Excluded ^a
Aslam 2019	USA	Case rpt.	1	MDR	Other	Combo	Pooled
Aslam 2019	USA	Case rpt.	1	MDR	IV	Combo	Pooled
Pirnay 2024	Belgium	Retro.	7	XDR	Other	Combo	Pooled
Pirnay 2024	Belgium	Retro.	23	MDR	Other	Combo	Pooled
Pirnay 2024	Belgium	Retro.	2	PDR	Other	Combo	Pooled
Köhler 2023	Switzerland	Case rpt.	1	MDR	Inhaled	Combo	Pooled
Chan 2025	USA	Case series	7	MDR	Inhaled	Combo	Pooled
Chan 2025	USA	Case series	2	PDR	Inhaled	Combo	Pooled
Law 2019	USA	Case rpt.	1	MDR	IV	Combo	Pooled
Hahn 2023	USA	Case series	2	MDR	Inhaled	Combo	Pooled
Levêque 2023	France	Case rpt.	1	XDR	Other	Combo	Pooled
Duplessis 2018	USA	Case rpt.	1	MDR	IV	Combo	Pooled
Denis 2026	France	Case rpt.	1	MDR	Other	Combo	Pooled
Malhotra 2026	USA	Case rpt.	1	MDR	NR	Combo	Pooled
Yang 2025	China	Case rpt.	1	MDR	Inhaled	Combo	Pooled
Li 2025	China	Case rpt.	1	MDR	Other	Combo	Pooled
Chan 2018	USA	Case rpt.	1	MDR	Topical	Combo	Pooled
Arya 2026	USA	Case rpt.	1	MDR	NR	Phage mono	Pooled
Ferry 2021	France	Case rpt.	1	NC	Topical	Combo	Pooled
Maddocks 2019	Australia	Case rpt.	1	NC	Other	Combo	Pooled
Aslam 2020	United States	Case series	1	MDR	IV	Combo	Pooled
Aslam 2020	United States	Case series	1	NC	IV	Combo	Pooled
Aslam 2020	United States	Case series	1	NC	IV	Combo	Pooled
Cesta 2023	Italy	Case rpt.	1	NC	Topical	Combo	Pooled
Khatami 2021	Australia	Case rpt.	1	NC	NR	Combo	Pooled
Zaldastanishvili 2021	Georgia	Case series	1	NC	Other	Phage mono	Pooled
Zaldastanishvili 2021	Georgia	Case series	1	NC	Other	Phage mono	Pooled
Green 2023	USA	Case series	1	NC	Other	Combo	Pooled
Casazza 2025	USA	Case rpt.	1	MDR	Other	Combo	Pooled
Teney 2024	France	Case rpt.	1	XDR	Other	Combo	Pooled
Li 2023	China	Case rpt.	1	MDR	Inhaled	Combo	Pooled
Chen 2022	China	Case rpt.	1	NC	NR	Combo	Pooled
Jennes 2017	Belgium	Case rpt.	1	XDR	Other	Phage mono	Excluded ^a
Chung 2026	Singapore	Case rpt.	1	<MDR	IV	Combo	Excluded ^a
Ronit 2024	Denmark	Case rpt.	1	<MDR	IV	Combo	Excluded ^a
Rubalskii 2020	Germany	Case series	1	NC	Topical	Combo	Pooled
Pardo-Freire 2025	Spain	Case rpt.	1	MDR	Inhaled	Combo	Pooled

Notes: One row per study-arm (the unit of extraction; Section 2.5), not per study – Liu, Lu, et al. (2025), Aslam, Courtwright, et al. (2019), and Chan et al. (2025) each contribute 2 arms. *n* is the arm-level patient count. Resistance: MDR/XDR/PDR per Magiorakos et al. (2012) where independently re-derivable from a reported antibiogram, else the primary study’s own label; NC = not-classifiable (no antibiogram published); <MDR = published antibiogram establishes non-susceptibility in fewer than the three categories the population criterion requires. Route and Modality are collapsed to their controlled category (parenthetical qualifiers, e.g. specific multi-route detail, are stripped; see `data/cleaned/phage_therapy_extraction_dataset.csv` and its codebook for the uncollapsed text). Pooling reflects this review’s pre-specified pooling-eligibility rule (Section 2.4), not risk of bias or study quality.

^a Leitner et al. (2021) is retained in the analytic dataset but excluded from all pooling: its outcomes are reported only as a six-pathogen trial-wide aggregate with no *Pseudomonas*-specific breakdown (Section 2.4).

Source: `scripts/R/09_table1_characteristics.R,` reading `data/cleaned/phage_therapy_extraction_dataset.csv.`

this is and is not. It is a systematic floor built only on facts the extraction already established, scoring a domain as unclear wherever the extraction is silent; it is not a substitute for a second reviewer reading each full text and judging each domain by hand, which this review has not had (Section 4.4).

The result is worse than the partial assessment this review previously reported, though no longer uniformly so. **38 of the 40 arms rate High risk of bias overall and 2 rate Moderate**, and the domain responsible is causality: 37 of the 40 arms administered phage together with antibiotics, so those arms cannot attribute their outcome to the phage rather than to the co-intervention. Causality is therefore High in 37 arms and Moderate in only 3—the three phage-monotherapy arms, where there is no co-intervention to separate the effect from. Two of those three rate Moderate *overall* (Zaldastanishvili et al.’s Eliava patients); Arya et al. (2026) still rates High, because publishing a single patient is itself the selection event. The other domains discriminate more sharply: selection is High in 25 arms, Moderate in 10 and Low in 5; ascertainment High in 2, Moderate in 26 and Low in 12; reporting High in 7, Moderate in 9 and Low in 24.

One quantity in that profile runs against intuition and is worth reporting. The 20 stand-alone single-patient reports — the arms rated High on selection, because publishing one patient is itself the selection event — together carry only 33 of the corpus’s 91 patients. Three larger sources (Pirnay et al., Onallah, Hazan, Nir-Paz, et al. and Chan et al. 2025) carry 56 between them. Since the pooling model weights by patient rather than by arm, the many case reports contribute a minority of the evidence in every cell: their share of patients ranges from 10% to 35% and never approaches half. The corpus is numerically dominated by a few series, which is the opposite of what its design distribution alone would suggest, and it is why the risk-of-bias downgrade in Section 3.8 is keyed to the uniform High overall rating rather than to the case-report share.

Two consequences follow, and neither favours this review. First, the GRADE risk-of-bias domain has stopped being *uniform*, and the history matters because it was the sharpest

methodological objection this review received. Through the previous round every arm rated High, so the domain deducted two levels everywhere and a referee objected — correctly — that a deduction following algebraically from a tautology is a constant dressed as an assessment. The two phage-monotheapy arms added at round 5 break the tautology: cells containing them now take -1 and the four MDR cells, which contain none, still take -2 (Table 8). The domain discriminates for the first time in this review, and it does so because the corpus acquired arms where causality is assessable, not because any threshold was tuned. Second, the earlier characterisation of the single-patient reports as “Low-to-moderate risk” was not supportable: it rested on a per-domain tabulation that was never retained, and the rule-based appraisal that replaces it reaches the opposite conclusion. These ratings feed the GRADE assessment in Section 3.8.

3.4 Synthesis of Pooled Outcomes

Of 48 outcome-stratum cells defined by the pre-specified four-way stratification, 13 met the threshold of at least 3 independent studies and 20 patients (Table 4); the remaining 35 are narrated in Table 5. Pooled clinical success across 31 studies (37 arms, 80 patients) was 78.7% (95% CI 68.0%–86.6%; Figure 2), directionally consistent with prior non-stratified syntheses (El Haddad et al., 2019; Uyttebroek et al., 2022; Liu, Thong, et al., 2025). It is explicitly secondary and contextual regardless, and one further qualification belongs with the number rather than in a later limitation: the sampling frame is the *published record*, so this is the proportion of published arms reporting success, not the proportion of treated patients achieving it, and under the favourable-outcome reporting this review argues is intrinsic to a compassionate-use case-report literature (Section 2.6) those two quantities differ by an unbounded amount. Pooled eradication across 27 studies (32 arms, 68 patients) was 50.4% (95% CI 29.2%–71.5%), pooled safety across 24 studies (30 arms, 60 patients) was 18.3% (95% CI 10.2%–30.8%), and pooled mortality across 32 studies (39 arms, 76 patients) was 9.2% (95% CI 4.3%–18.5%). A property of these intervals must be stated plainly because

Table 2: Per-arm risk of bias, all 40 pooling-eligible study-arms

Study-arm	Design	<i>N</i>	Sel.	Asc.	Cau.	Rep.	Overall
Tkhilaishvili2020_A	case report	1	H	L	H	L	H
Blasco2023_A	case report	1	H	L	H	L	H
Ngaury2026_A	case report	1	H	L	H	L	H
Liu2025_perinephric_P2	case series	1	M	L	H	L	H
Racenis2022_femur_A	case report	1	H	L	H	L	H
Onallah2023_PASA16_agg	case series	15	M	M	H	H	H
Aslam2019_P1	case report	1	H	M	H	L	H
Aslam2019_P2	case report	1	H	M	H	L	H
Pirnay2024_XDR	retrospective cohort	7	L	L	H	L	H
Pirnay2024_MDR	retrospective cohort	23	L	L	H	L	H
Pirnay2024_PDR	retrospective cohort	2	L	L	H	L	H
Kohler2023_A	case report	1	H	M	H	L	H
Chan2025_MDR	case series	7	L	M	H	H	H
Chan2025_PDR	case series	2	L	M	H	H	H
Law2019_A	case report	1	H	M	H	L	H
Hahn2023_A	case series	2	M	M	H	H	H
Leveque2023_A	case report	1	H	M	H	L	H
Duplessis2018_A	case report	1	H	M	H	H	H
Denis2026_A	case report	1	H	M	H	L	H
Malhotra2026_A	case report	1	H	M	H	H	H
Yang2025_A	case report	1	H	M	H	L	H
Li2025_biliary_A	case report	1	H	H	H	L	H
Chan2018_omko1_A	case report	1	H	M	H	L	H
Arya2026_pji_A	case report	1	H	M	M	H	H
Ferry2021_A	case report	1	H	M	H	M	H
Maddocks2019_A	case report	1	H	H	H	M	H
Aslam2020_C5	case series	1	M	L	H	M	H
Aslam2020_C8	case series	1	M	L	H	M	H
Aslam2020_C10	case series	1	M	L	H	M	H
Cesta2023_A	case report	1	H	M	H	M	H
Khatami2021_A	case report	1	H	M	H	M	H
Zaldastanishvili2021_P1	case series	1	M	M	M	L	M
Zaldastanishvili2021_P2	case series	1	M	M	M	L	M
Green2023_C11	case series	1	M	M	H	L	H
Casazza2025_A	case report	1	H	M	H	M	H
Teney2024_A	case report	1	H	M	H	L	H
Li2023_ILD_A	case report	1	H	M	H	L	H
Chen2022_empyema_A	case report	1	H	M	H	M	H
Rubalskii2020_P8	case series	1	M	L	H	L	H
PardoFreire2025_A	case report	1	H	M	H	L	H

Notes: Domains are those of Murad et al. (2018) for case reports and series: Sel. = selection, Asc. = ascertainment, Cau. = causality, Rep. = reporting. L = Low, M = Moderate, H = High. Overall is the worst domain, stated as the rule rather than left implicit; Murad’s tool is not additive and offers no summary score. Every arm rates High overall because causality cannot be satisfied wherever phage was co-administered with antibiotics (37 of 40 arms). Scores are generated by pre-specified rules over recorded extraction metadata (`scripts/R/12_risk_of_bias.R`), which makes the appraisal reproducible and auditable but does not substitute for a second reviewer judging each domain against each full text. Source: `scripts/R/12_risk_of_bias.R`.

it is easy to miss: τ^2 is estimated at exactly zero in 11 of the 13 pooled cells, and where that happens the binomial-normal GLMM with a logit link returns the crude event fraction itself — 78.7% is 63/80, 18.3% is 11/60, 9.2% is 7/76— with an interval within rounding of an exact binomial interval. With 33 of 40 arms contributing a single patient, a random intercept is identified only by the 7 multi-patient arms, so $\tau^2 = 0$ here is *non-identification of between-arm variance, not evidence of homogeneity*, and the prediction interval collapses onto the confidence interval in those same cells. The most consequential of those defects is now auditable rather than asserted. Table 3 lists every pooling-eligible arm’s clinical-success definition exactly as recorded from its source, with the longest follow-up horizon that source states. Only 15 of the 40 arms state a horizon at all; the other 25 report success without saying over what window it was observed. Among those that do, the range runs from 3 days to 2 years, a 244-fold spread. At one end, Maddocks et al. (2019) counts improvement of oxygenation with successful ventilator weaning; at the other, Arya et al. (2026) and Cesta et al. (2023) count absence of relapse two years after therapy stopped. Both enter the same numerator, weighted identically. We do not think a single percentage should be read across that range, and the table is provided so a reader can decide for themselves which arms they are willing to pool. The table also records what this review still cannot supply: the extraction schema has no syndrome field, and deriving one by keyword from the recorded text left a quarter of the arms unclassified and misassigned others, so a syndrome-stratified reading is not available here (Section 4.4).

Two of these four constructs are less coherent than a single pooled percentage makes them look, and we state the defect rather than leaving it to be inferred. *Mortality has no common follow-up horizon*. The 7 deaths among 76 patients accrue over windows ranging from roughly two weeks to twenty-four months, and 6 of the 7 are attributed by their own source authors to something other than the phage-treated infection: Ngauiy et al. (2026)’s patient died about two weeks after discharge of failure to thrive with a culture-negative implant; Levêque et al. (2023)’s died at twenty-four months of multi-organ failure *after* microbiological

Table 3: Clinical-success definitions and follow-up horizons, by study-arm

Study-arm	<i>n</i>	Follow-up	Clinical-success definition as recorded from the source
Maddocks2019_A	1	3 d	Improvement of oxygenation within 3 days with successful ventilator weaning, in a 77-year-old woman with VAP and empyema complicated by bronchopleural fistula after thoracotomy
Blasco2023_A	1	7 d	Treatment FAILURE (paper title): new <i>P. aeruginosa</i> BSI with septic residual limb metastasis occurred in absence of antimicrobial therapy after the 1-week phage course; ultimately required vascular prosthesis replacement
Ngaury2026_A	1	2 wk	Clinical and microbiological resolution of hypermutator XDR osteomyelitis/hardware infection; ESR/CRP normalized after 1 week; orbital implant culture-negative at removal 2 weeks post-phage
Yang2025_A	1	2 wk	Author-reported significant clinical improvement after 2 weeks of nebulized phage therapy (vasopressors discontinued, haemodynamics stable, inflammatory markers and lung effusion decreased).
Aslam2020_C10	1	3 mo	Recurrent <i>P. aeruginosa</i> bacteremia from a probable aortic graft infection; blood cultures negative on therapy and four weekly surveillance cultures negative with no recurrence through 12 weeks after end of therapy, against a pre-treatment pattern of recurrence 7-10 days off antibiotics
Teney2024_A	1	7 mo	52-year-old man with self-inflicted burns over 81% of total body surface area and four successive episodes of ventilator-associated pneumonia with bacteraemia from an NDM-1-producing XDR strain. Success recorded as reduced bronchorrhoea and oxygen dependence, improving skin grafts, extubation on the last day of the second phage course, and discharge from intensive care after 203 days
Racenis2022_femur_A	1	9 mo	Eradication of infection in PROXIMAL femoral segment/hip region (site treated with local phage BFC 1.10); wound healed with no local/systemic infection signs at days 1-15 post-phage; enabled endoprosthesis reconstruction 9 months later. IMPORTANT: a SEPARATE, untreated-by-phage recurrence of MDR <i>P. aeruginosa</i> +VRE occurred at the DISTAL femoral segment 9 months later and was treated with antibiotics alone (DAIR, colistin+fosfomycin) - that distal episode is NOT part of this phage-treated arm and is excluded from this row's outcome counts
Law2019_A	1	9 mo	Author-reported clinical resolution of MDR <i>P. aeruginosa</i> pneumonia; no recurrence of pseudomonal pneumonia or CF exacerbation within 100 days of ending bacteriophage therapy; successful bilateral lung transplantation 9 months later.
Tkhilaishvili2020_A	1	10 mo	Infection eradicated; no pain, satisfactory mobility, normal CRP, no implant loosening on X-ray at 10-month follow-up
Ferry2021_A	1	12 mo	Resolution of relapsing prosthetic knee infection: normal local knee status, painless motion and walking at 1-year follow-up, on suppressive antimicrobial therapy
Chan2018_omko1_A	1	18 mo	Chronic MDR <i>P. aeruginosa</i> aortic Dacron graft infection resolved after a single local application of phage OMKO1 + cef-tazidime; no evidence of recurrent infection over 18 months off all antibiotics (blood cultures, clinical symptoms, CT).
Denis2026_A	1	2.0 yr	Author-reported cure: at 2-year follow-up the patient had no relapse of the previously thrice-relapsing MDR <i>P. aeruginosa</i> frontal osteitis, without any suppressive treatment.
Arya2026_pji_A	1	2.0 yr	Author-reported clinical resolution of a multidrug-resistant, non-operable <i>P. aeruginosa</i> periprosthetic joint infection that had failed multiple antibiotic and surgical interventions, achieved with intermittent bacteriophage therapy alone (no antibiotics) over 2 years; microbiological eradication was NOT achieved (chronic suppression, altered virulence, biofilm disruption).
Cesta2023_A	1	2.0 yr	Chronic hip prosthetic joint infection: at 2-year follow-up after suspension of therapy there were no clinical signs of relapse and a labelled-leukocyte scan showed no pathological uptake
Leveque2023_A	1	2.0 yr	Clinical failure: despite microbiological eradication of the XDR strain, the patient progressed to multi-organ failure and died at 24 months post-lung-transplant; the chronic bronchopulmonary infection favored the fatal course.
Liu2025_perinephric_P2	1	not stated	Treatment FAILURE for primary abscess site: urinary tract infection resolved and inflammatory area reduced (72x15mm to 21x10mm on ultrasound) but perinephric <i>P. aeruginosa</i> PERSISTED in wound secretions despite 3 courses of phage therapy; patient discharged per personal choice, not cured
Onallah2023_PASA16_15gg	1	not stated	'Good clinical outcome' as defined by study authors (binary); 2 clinical failures among 15

eradication; and all four deaths inside Pirnay et al.’s cohort are coded by that study as unlikely related to phage therapy (septic shock, tumour progression, postoperative ileus). The pooled 9.2% is therefore an all-cause proportion over an unstandardised window, not an infection-attributable mortality rate, and it should not be read as the latter. *Safety mixes two denominators*. Some arms record *any* adverse event and others only events the source attributed to phage therapy; where a source reported only that no *severe* event occurred (Cesta et al. (2023)), the numerator is left missing rather than set to zero, because a severe-event denominator cannot be folded into an any-event pool. The direction of the resulting bias is downward and its size is not estimable from the published reports. The eradication estimate carries the only substantial between-arm variance in the review ($\tau^2 = 1.995$), and an earlier version of this section read that width as “a genuine feature of the clinical spectrum, not as noise”. That reading is wrong, and the test that shows it is simple enough to state in a sentence. Eradication means different things at different sites. In *established* chronic airway infection — cystic fibrosis, primary ciliary dyskinesia with bronchiectasis, chronic interstitial-lung-disease colonisation — clearance is neither achieved nor the therapeutic target; the endpoints that field uses are bacterial density, exacerbation frequency and lung function, and eradication is a concept reserved for early acquisition rather than for the established biofilm state. Elsewhere — a sterilised bacteraemia, an explanted culture-negative implant, a healed surgical wound — clearance is exactly the goal.

Splitting the corpus on that line resolves the variance completely. Among the 24 arms at sites other than a chronically colonised airway (21 studies, 53 patients, 33 eradications), the pooled proportion is 62.3% (95% CI 47.9%–74.8%) with $\tau^2 = 0.000$ — *exactly zero*. Among the 8 chronic-airway arms (6 studies, 15 patients, only 2 eradications), the model degenerates: 3.1% with an interval of 95% CI 0.0%–96.9% and $\tau^2 = 17.097$, which is the boundary behaviour of a near-zero-event cell and not an estimate of anything.

So the heterogeneity was never a property of phage therapy. It was produced by pooling two endpoints that share a name, and the pooled 50.4% is a weighted average of a coherent

62% and a degenerate 3% whose mixing proportion is a fact about which patients happened to be published. We report the overall figure because it is what the pre-specified analysis produces, but a reader should use the split, and no reader should read the overall interval as clinical variation. The same objection applies with less force to the other three outcomes, whose constructs are described in Table 3; eradication is where it is decisive because the endpoint is not merely defined differently across arms but is not the same clinical event.

Table 4: Pooled outcome-stratum cells meeting the pre-specified pooling threshold

Outcome – Stratum	k	Arms	N	Pooled proportion [95% CI]	τ^2	95% prediction interval	Model
Clinical success – Overall	31	37	80	78.7% [68.0%, 86.6%]	0.000	[68.0%, 86.6%]	GLMM
Safety (≥ 1 AE) – Overall	24	30	60	18.3% [10.2%, 30.8%]	0.000	[10.2%, 30.8%]	GLMM
Eradication – Overall	27	32	68	50.4% [29.2%, 71.5%]	1.995	[4.7%, 95.4%]	GLMM
Mortality – Overall	32	39	76	9.2% [4.3%, 18.5%]	0.000	[4.3%, 18.5%]	GLMM
Clinical success – Resistance: MDR	17	18	40	75.0% [58.1%, 86.6%]	0.000	[58.1%, 86.6%]	GLMM
Clinical success – Resistance: NC	10	12	26	88.5% [66.5%, 96.7%]	0.000	[66.5%, 96.7%]	GLMM
Safety (≥ 1 AE) – Resistance: MDR	14	15	38	18.4% [8.4%, 35.7%]	0.000	[8.4%, 35.7%]	GLMM
Eradication – Resistance: MDR	16	17	45	46.5% [12.0%, 84.7%]	6.092	[0.3%, 99.6%]	GLMM [‡]
Mortality – Resistance: MDR	19	20	49	6.1% [1.8%, 18.5%]	0.000	[1.8%, 18.5%]	GLMM
Clinical success – Route: other	11	14	57	78.9% [65.0%, 88.3%]	0.000	[65.0%, 88.3%]	GLMM
Safety (≥ 1 AE) – Route: other	8	11	40	15.0% [6.2%, 32.1%]	0.000	[6.2%, 32.1%]	GLMM
Eradication – Route: other	10	13	42	59.5% [42.6%, 74.5%]	0.000	[42.6%, 74.5%]	GLMM
Mortality – Route: other	10	13	42	11.9% [4.6%, 27.6%]	0.000	[4.6%, 27.6%]	GLMM

Notes: Sample: 40 pooling-eligible study-arms across 33 studies, 91 patients (`data/cleaned/phage_therapy_extraction_dataset.csv`). k is the number of distinct studies, “Arms” the number of contributing study-arms, and N the pooled patient count. Both k and Arms are reported because the model is fitted at the arm level: a study contributing more than one arm (for example Pirnay et al.’s three resistance-stratified arms) enters the likelihood once per arm, so the Hartung-Knapp-Sidik-Jonkman interval uses one fewer degree of freedom than the arm count, not the study count. Reporting only k would leave the intervals irreproducible. NC = not-classifiable (no antibiogram published); <MDR = published antibiogram establishes non-susceptibility in fewer than the three categories the population criterion requires. [‡] marks a cell whose 95% confidence interval spans at least 50 percentage points: such a cell is retained because it is the only estimate its stratum has, but it is as uninformative about the underlying proportion as the zero-event cells demoted to Table 5, and must not be read as an estimate. A cell is reported here only if it contains at least 3 independent studies and at least 20 pooled patients; all other cells appear in Table 5. Pooled proportion, τ^2 , and the 95% prediction interval are from a random-effects binomial-normal GLMM (logit link), with 95% CIs computed under a requested Hartung-Knapp-Sidik-Jonkman adjustment that does not bind in 11 of these 13 cells; in those the interval reduces to a t -interval on the logit of the pooled counts (Section 3.6). Between-study heterogeneity is summarized by τ^2 and the 95% prediction interval, which for a binomial-normal GLMM are more reliable indices for proportion meta-analyses than the conventional inconsistency statistic. “Overall” rows are secondary/contextual and are not this review’s primary estimand. Source: `scripts/R/04_estimation.R`, `08_tables.R`.

Four stratified cells reached the threshold, and two of the four are residual categories

rather than clinical classes — a fact that shapes how much any of this can be relied on. The clearest positive result is a genuine resistance class: the **MDR stratum is pooled across all four outcomes** — clinical success 75.0% (95% CI 58.1%–86.6%) (17 studies, 40 patients; Figure 3), safety 18.4% (95% CI 8.4%–35.7%) (14 studies, 38 patients), eradication 46.5% (95% CI 12.0%–84.7%) (16 studies, 45 patients), and mortality 6.1% (95% CI 1.8%–18.5%) (19 studies, 49 patients). This answers part of the review’s originally-blocked core stratification aim — an MDR-specific pooled estimate now exists — but only as a fragile, single-cohort signal: Pirnay et al. still supplies roughly three-fifths of the MDR clinical-success patients, and the estimate clears the threshold only under author-reported resistance labels, collapsing to three studies and four patients when classification is restricted to independently verified cases (Section 3.6).

It does not complete the aim as designed. **XDR (5 studies, 11 patients) and PDR (2 studies, 3 patients) remain below the 20-patient threshold for every outcome** and are reported in Table 5 as insufficient evidence, so the MDR-*versus*-XDR comparison this review was designed to make still cannot be made, even though MDR alone can now be estimated.

Exactly one further resistance cell pools, and it is the most uncomfortable number in this review. Clinical success in the not-classifiable stratum is 88.5% (95% CI 66.5%–96.7%) across 10 studies and 26 patients — the *highest* point estimate anywhere in the review, in the one stratum defined by the *absence* of resistance documentation. We do not read this as evidence that undocumented infections respond better. Two readings cannot be separated with these data: reports that omit an antibiogram may be systematically the ones with favourable outcomes, or the stratum may simply contain less-resistant patients whom the eligibility rule admitted. The first is reporting bias and the second is a defect in this review’s own inclusion criterion; neither is reassuring, and Section 4.4 treats them as the review’s central threat rather than as a caveat.

The remaining two not-classifiable cells do *not* pool. Safety reaches only 6 studies and 8

patients; mortality 9 studies and 11 patients. Earlier drafts of this section reported both as pooled, with point estimates and confidence intervals. Those cells never met the threshold and those numbers had no counterpart in any pipeline output. They are withdrawn, and Section 4.4 records how they survived four rounds of review, because that failure is itself a finding about this manuscript.

On the route axis, the multi-route or otherwise unclassifiable (“other”) category is pooled across all four outcomes: clinical success 78.9% (95% CI 65.0%–88.3%) (11 studies, 57 patients), safety 15.0% (95% CI 6.2%–32.1%) (8 studies, 40 patients), eradication 59.5% (95% CI 42.6%–74.5%) (10 studies, 42 patients), and mortality 11.9% (95% CI 4.6%–27.6%) (10 studies, 42 patients). **No specific single route reaches the threshold for any outcome.** Inhaled/nebulized comes closest on study count and still falls far short on patients: its safety cell holds 5 studies and 6 patients. Where a route records no deaths at all we report an exact one-sided (Clopper-Pearson) bound rather than a pooled zero, since a pooled 0.0% with an interval spanning the whole range conveys nothing: inhaled/nebulized has 0 deaths among 15 patients (97.5% upper bound 21.8%), and topical/local 0 among 7 (upper bound 41.0%). Both bounds are consistent with mortality as high as roughly one patient in four, and neither supports any claim of route-specific safety.

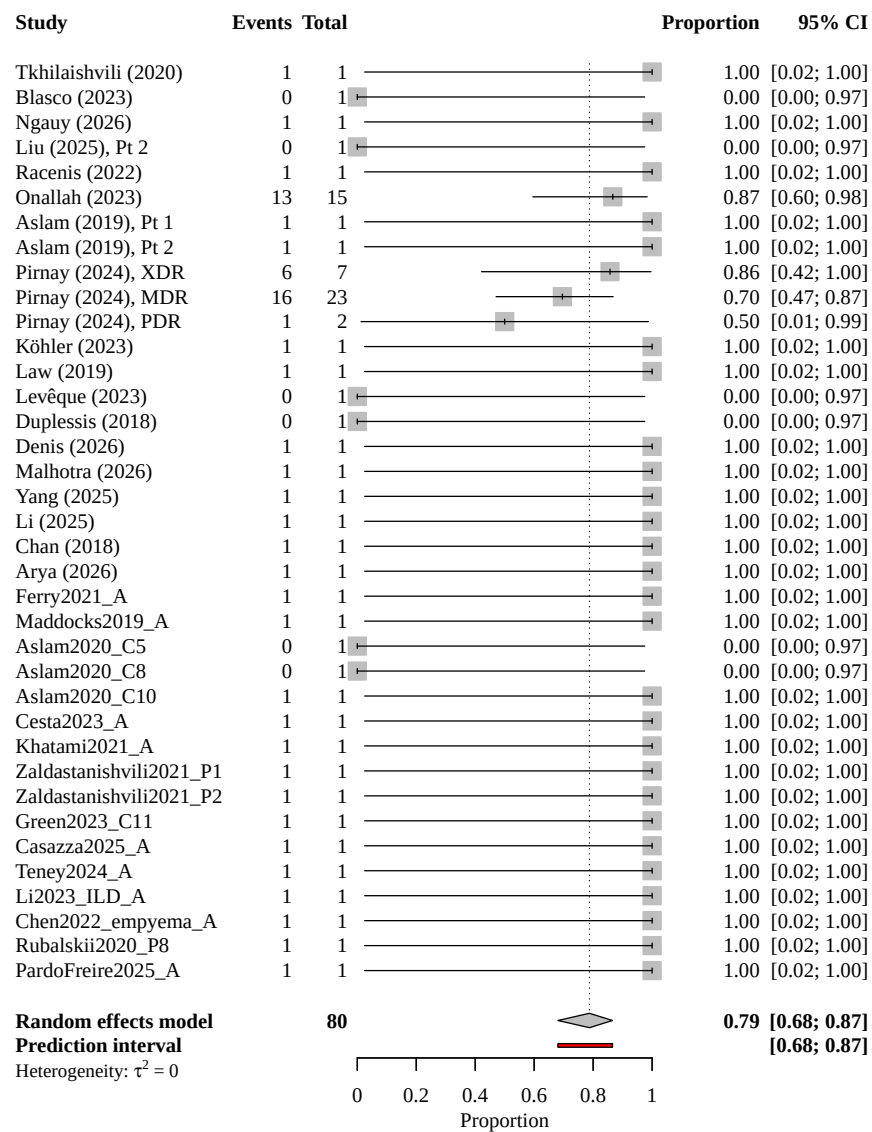
Modality still could not be stratified. 3 of the 40 arms are phage monotherapy and the other 37 combine phage with antibiotics, so there is no antibiotic-monotherapy stratum to compare against and the phage-monotherapy stratum holds three patients — far below the twenty-patient floor (Table 5, final row). The direction of the point stands and is the cleanest statement of what this corpus cannot answer: in all but 3 arms the published record does not contain a patient given phage without a concomitant antibiotic, so almost no arm here can attribute its outcome to the phage rather than to the co-intervention. Those three arms are correspondingly the most informative in the review on that question, and two of the three — Zaldastanishvili et al.’s Eliava patients — did *not* achieve eradication.

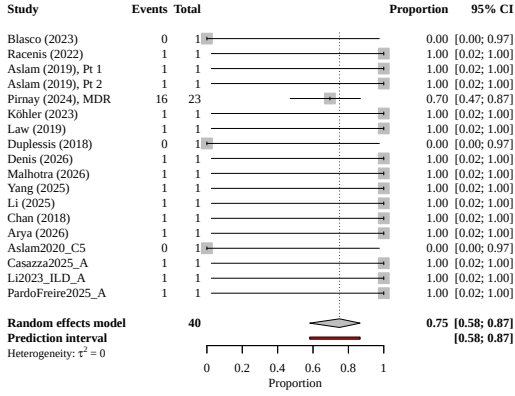
Pooled safety (at least one adverse event), across 24 studies (30 arms, 60 patients), is

Table 5: Outcome-stratum cells reported narratively: insufficient evidence to support formal pooling

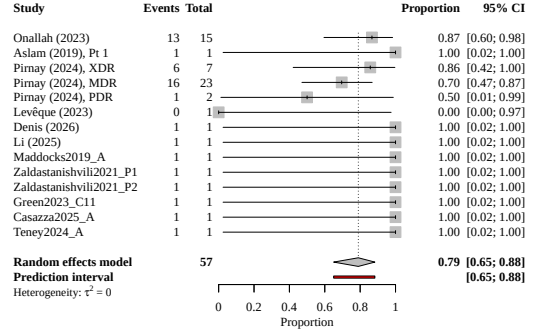
Outcome – Stratum	k	Arms	N	Reason not pooled
Clinical success – Resistance: PDR	2	2	3	$k < 3, N < 20$
Clinical success – Resistance: XDR	5	5	11	$N < 20$
Safety (≥ 1 AE) – Resistance: NC	6	8	8	$N < 20$
Safety (≥ 1 AE) – Resistance: PDR	2	2	3	$k < 3, N < 20$
Safety (≥ 1 AE) – Resistance: XDR	5	5	11	$N < 20$
Eradication – Resistance: NC	6	7	7	$N < 20$
Eradication – Resistance: PDR	3	3	5	$N < 20$
Eradication – Resistance: XDR	5	5	11	$N < 20$
Mortality – Resistance: NC	9	11	11	$N < 20$
Mortality – Resistance: PDR	3	3	5	$N < 20$
Mortality – Resistance: XDR	5	5	11	$N < 20$
Clinical success – DTR: positive	4	4	5	$N < 20$
Clinical success – DTR: negative	1	1	1	$k < 3, N < 20$
Clinical success – DTR: not derivable	28	32	74	partial antibiogram
Safety (≥ 1 AE) – DTR: positive	4	4	5	$N < 20$
Safety (≥ 1 AE) – DTR: negative	1	1	1	$k < 3, N < 20$
Safety (≥ 1 AE) – DTR: not derivable	21	25	54	partial antibiogram
Eradication – DTR: positive	5	5	7	$N < 20$
Eradication – DTR: negative	1	1	1	$k < 3, N < 20$
Eradication – DTR: not derivable	24	26	60	partial antibiogram
Mortality – DTR: positive	5	5	7	$N < 20$
Mortality – DTR: negative	1	1	1	$k < 3, N < 20$
Mortality – DTR: not derivable	29	33	68	partial antibiogram
Clinical success – Route: inhaled/nebulized	4	4	4	$N < 20$
Clinical success – Route: IV	6	8	8	$N < 20$
Clinical success – Route: topical/local	7	7	7	$N < 20$
Safety (≥ 1 AE) – Route: inhaled/nebulized	5	5	6	$N < 20$
Safety (≥ 1 AE) – Route: IV	5	7	7	$N < 20$
Safety (≥ 1 AE) – Route: topical/local	5	5	5	$N < 20$
Eradication – Route: inhaled/nebulized	5	6	13	$N < 20$
Eradication – Route: IV	5	5	5	$N < 20$
Eradication – Route: topical/local	6	6	6	$N < 20$
Mortality – Route: inhaled/nebulized	6	7	15	$N < 20$
Mortality – Route: IV	6	8	8	$N < 20$
Mortality – Route: topical/local	7	7	7	$N < 20$
Modality: antibiotic monotherapy (all outcomes)	0	0	0	0 arms (stratum empty)

Notes: Sample: 40 pooling-eligible study-arms across 33 studies. NC = not-classifiable (no antibiogram published); <MDR = published antibiogram establishes non-susceptibility in fewer than the three categories the population criterion requires. The 21 outcome-stratum cells listed here (plus the final, empty antibiotic-monotherapy row, giving 22 rows in total) either fall short of the pre-specified threshold ($k \geq 3$ studies, $N \geq 20$ patients) on at least one criterion or are uninformative as a pooled proportion; "Reason not pooled" states which applies for that cell ($k < 3$: too few studies; $N < 20$: too few patients despite enough studies; both may fail together; "0 events": a zero-event cell that meets the count threshold but yields an uninformative pooled proportion, reported narratively instead). Individual study-level proportions are tabulated in the underlying analysis but not pooled, per the pre-registered sparsity rule (Section 2.7). This includes XDR and PDR for every outcome (failing on N alone – both meet the study-count threshold), every specific single route apart from the pooled "other" category and the newly-pooled inhaled/nebulized safety cell, and the inhaled/nebulized mortality cell, which meets the count threshold but is demoted for reporting zero events (Section 3.4). The final row reports the absent antibiotic-monotherapy stratum across all four outcomes. Source: `scripts/R/04_estimation.R`, `08_tables.R`.





(a) MDR resistance stratum



(b) Route: “other” (multi-route)

Figure 3: Forest plots: two stratified cells meeting the pooling threshold. Each row is one study-arm, not one study, so the row count exceeds the study count where a study contributes several arms. Panel (a): clinical success in the MDR stratum — 18 arms from 17 studies, 40 patients, pooled 75.0% (95% CI 58.1%–86.6%). This estimate clears the pooling threshold only under author-reported resistance labels; restricting to independently verified classification collapses it below threshold (Section 3.6). Panel (b): clinical success among arms coded a multi-route or otherwise unclassifiable (“other”) route — 14 arms from 11 studies, 57 patients, pooled 78.9% (95% CI 65.0%–88.3%). Both cells have $\tau^2 = 0$, so the plotted interval is a t -interval on the logit of the pooled counts and carries no between-arm variance component (Section 3.6). XDR, PDR and every specific single route fell below threshold and are not plotted; see Table 5. Source: `scripts/R/07_figures.R`.

18.3% (95% CI 10.2%–30.8%; $\tau^2 = 0.000$; Figure 4). Leave-one-out does not flag the *overall* safety estimate as sensitive to any single study’s removal, though the MDR-specific safety estimate, the mortality-specific estimate in both the MDR and not-classifiable resistance strata, and the route-“other” safety estimate remain sensitive to removing Pirnay et al. (MDR, route-“other”) or Jault et al. (not-classifiable mortality) specifically, collapsing to a value indistinguishable from zero on removal. The sensitivity models bracket the GLMM: Freeman-Tukey gives 7.8% and logit-DerSimonian-Laird gives 28.0% against the GLMM’s 18.3% (Table 6), and the fixed-effect model converges at 28.0%. The Freeman-Tukey value lies *outside* the primary interval, which is reported here rather than left to the table: an interval that excludes the estimate an equally defensible transformation returns on the same data is not a 95% interval for any quantity a reader can use. Even so, the MDR- and route-specific safety/mortality estimates, and the two not-classifiable safety/mortality cells, should be read as resting heavily on one or two dominant sources, not as an even-weighted synthesis across independent studies (Section 3.6).

One pooled cell requires explicit comment because it is both the highest estimate in the analysis and the least defensible. Clinical success in the *not-classifiable* resistance stratum is 88.5% (95% CI 66.5%–96.7%) (10 studies, 12 arms, 26 patients) — higher than the overall figure and higher than the MDR estimate. We report it here rather than leave it in Table 4 unmentioned, and we state why it should carry no weight: this stratum is defined by the *absence* of resistance data, so it is not a clinical population at all. Its patients may or may not have had resistant infection; what they have in common is only that their reports did not say. That the highest success proportion in the review sits precisely where resistance was never documented is worth stating plainly, because the most economical explanation is not that undocumented infections respond better but that reports which omit a susceptibility panel differ systematically from those that include one — most likely in how much detail a favourable outcome invited. It is a signal about reporting, not about phage therapy, and it is the strongest internal evidence in this review for the publication-bias downgrade applied

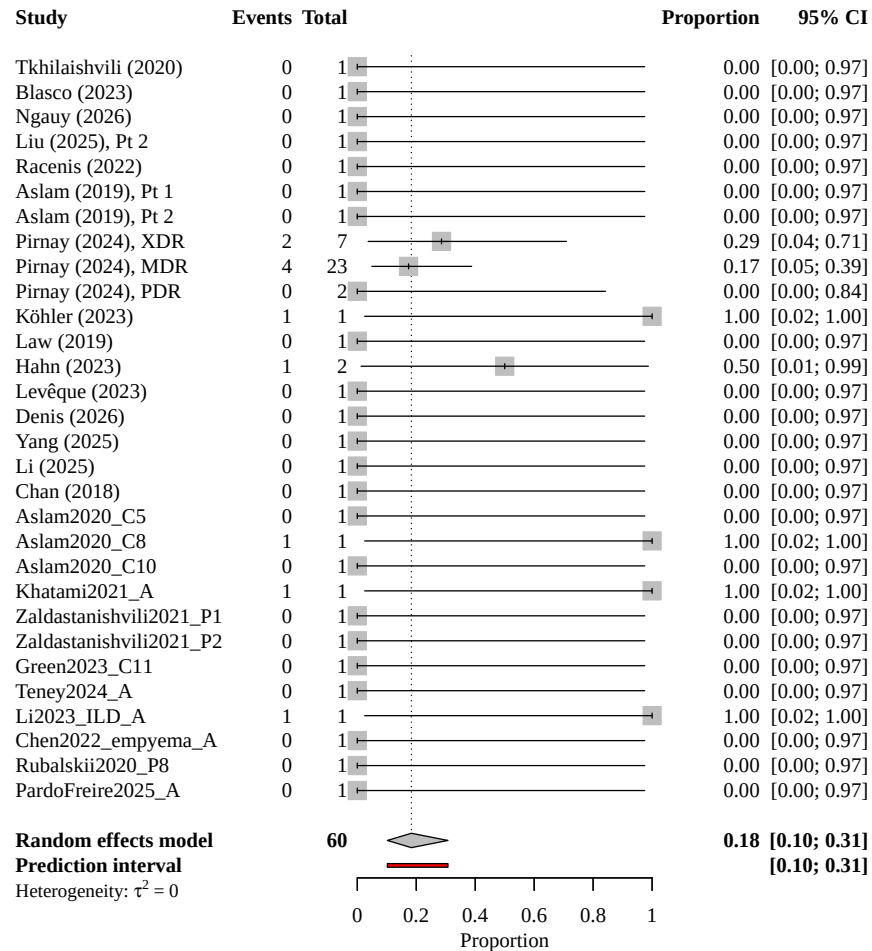


Figure 4: Forest plot: pooled safety (at least one adverse event), overall (secondary/contextual). Each row is one of the 30 study-arms contributing safety data (24 studies, 60 patients); the GLMM diamond shows 18.3% (95% CI 10.2%–30.8%). Leave-one-out (Section 3.6) no longer flags this specific estimate as sensitive to any single study’s removal, though the MDR- and route-specific

to every GRADE cell.

One stratum fails for a reason none of the others do, and it is the one a clinician would most want. Difficult-to-treat resistance is adjudicable for **5 arms positive and 1 negative, 6 of 40 in all**; the remaining 34 cannot be called in either direction. The positive cells carry too few patients to pool and the single negative arm carries one, so no DTR cell reaches the threshold (Table 5).

The reason those 34 arms cannot be classified is worth stating precisely, because an earlier version of this review stated it wrongly. It is *not* that their sources publish no antibiogram — several publish a panel, and this review’s own extraction records the agents. It is that the panels are *partial*: Kadri et al. (2018)’s criterion requires knowing the status of every first-line category, and a single untested category blocks both a positive and a negative call. Sources typically name the agents that were administered rather than the agents that were tested, which is a different and smaller set. Where a source does report enough — an isolate susceptible only to colistin, or only to ceftazidime-avibactam, or documented susceptible to ciprofloxacin — the call is immediate.

Earlier rounds of this review reported that DTR “cannot be computed from the published record at any sample size”. That claim was an artefact of this review’s own criterion, which required non-susceptibility to *all* β -lactams rather than to Kadri et al.’s first-line set, and was therefore stricter than the definition it cited (Section 2.2). Correcting it moved six arms and made DTR-negative reachable at all, a level the previous criterion had rendered impossible — which, in retrospect, was the diagnostic sign that something was wrong. The weaker claim that survives is still worth making: the comparison current guidance treats as decision-relevant for *P. aeruginosa* is reported too rarely and too incompletely to stratify a synthesis of this literature, and that is a statement about reporting practice rather than about phage therapy.

The same failure recurs on the other side of the comparison, and there it is sharper, because it concerns not a covariate but whether the intervention was delivered. Of the 40

pooling-eligible arms, only 3 record any phage-susceptibility testing at all — a phagogram, an efficiency-of-plating measurement, a spot test, or a plain statement that the preparation was active on the patient’s isolate. For the remaining 37 arms, 92% of the corpus, the published record does not establish that the administered phage could lyse the organism it was given for. This review is not entitled to assume it did: the assumption is exactly what Onsea et al. (2019) falsifies, an arm in which the cocktail was administered and the isolate was non-susceptible to it. Having excluded that arm on precisely this ground (Section 2.2), the review treats in-vitro activity as decision-relevant while being able to verify it in a small minority of the arms it retains. We resolve the asymmetry in the direction that costs us rather than the one that flatters the estimate: the rule stands, and the unverifiable majority is carried into the certainty assessment as indirectness (Section 3.8) instead of being silently counted as treated. A pooled proportion cannot distinguish a phage that worked from a phage that was never active, and for most of this corpus neither can the source.

3.5 Secondary Outcomes Reported Narratively

Two pre-specified secondary outcomes could not be pooled and are reported here rather than omitted, since silently dropping a pre-specified outcome is precisely the selective-reporting behaviour this review criticises in its source literature.

Emergence of phage resistance during treatment. 9 of the 40 pooling-eligible arms assess this outcome, and 4 of those 9— 44%— document phage resistance emerging during therapy. No pooled proportion is defensible from a denominator of 9, and none is offered, but the ratio should not be buried: *where investigators looked, they found it about half the time*. The documented mechanisms differ and are worth separating. Li et al. (2025) records O-antigen-deficient escape mutants arising under a four-phage cocktail, forcing a switch to a double-stranded RNA phage — receptor loss, the canonical route. Zaldastanishvili et al. (2021) documents population diversification by pulsed-field gel electrophoresis across serial isolates in both patients, with susceptibility differing between co-existing strains rather than

being lost outright. Yang et al. (2025) records emergence as the reason the bacterial load fell without the organism being eradicated. Running the other way, Chan, Turner, et al. (2018) is the case in which phage pressure *restored* antibiotic susceptibility through the efflux-pump trade-off that motivates phage steering.

The reporting rate remains the second finding rather than the first: 31 of 40 arms are silent on whether the organism became phage-resistant, so this review cannot say how often it happens — only that among the minority who checked, it happened often. An earlier version of this section reported one event among four arms and treated the reporting rate as the whole story. That understated the events by a factor of four, and one of the four (Yang et al.) had been coded as not assessed although its own source describes the emergence explicitly.

Length of hospitalization. Reported by 0 of the 40 pooling-eligible arms, so no synthesis was attempted (Section 2.7). Earlier drafts reported this as one arm. The single length-of-stay value in the extraction dataset belongs to Racenis et al., which the de-duplication rule removes from pooling, so no arm contributing to any estimate in this review reports it.

3.6 Sensitivity and Robustness Analyses

Transformation and model choice changes the pooled estimate by more than 5 percentage points in 12 of the 13 pooled cells; the three cells below the flag threshold are eradication overall, not-classifiable mortality, and the inhaled/nebulized safety cell (Table 6). Leave-one-out removal of each (arm, cell) combination shifts a pooled estimate outside its full-sample CI in exactly 4 cases: removing Pirnay et al. (2024) collapses the MDR-specific safety and mortality estimates and the route-”other” safety estimate to a value numerically indistinguishable from zero, and removing Jault et al. (2019) does the same to the not-classifiable resistance stratum’s new mortality estimate. Notably, the *overall* safety and mortality estimates are not shifted outside their CIs by any single-study removal. The same removals do *not* overturn the clinical-success or eradication estimates in any stratum, including MDR,

where Pirnay supplies the large majority of pooled patients (Section 3.4) — the instability is confined to the safety/mortality outcomes in specific strata rather than a general fragility of the MDR or not-classifiable pools. The Hartung-Knapp-Sidik-Jonkman adjustment is requested for every cell, and a diagnostic reports what it actually does. Comparing each cell’s model standard error against the naive complete-pooling binomial standard error on the logit scale, the ratio is exactly 1.0000 in **11 of the 13 pooled cells**. In those cells the adjustment does not bind at all: the reported interval is $\text{logit}(r/N) \pm t_{k-1} \sqrt{1/r + 1/(N - r)}$, a t -interval computed from the pooled counts, and the only respect in which it exceeds a Wald interval is the t quantile in place of z (2.028 against 1.960 at 36 degrees of freedom). It is not a variance-inflated random-effects interval and we do not describe it as one. The two exceptions are the cells with $\tau^2 > 0$ — eradication overall and eradication within MDR — where the ratio reaches 2.92 and the adjustment does substantial work. Two consequences follow. First, the reported precision in 11 cells is that of a single pooled binomial sample, which assumes every patient contributes independently; the cluster-robust check below is the only thing in this review that interrogates that assumption, and it does not bound these intervals. Second, any reading of these intervals as reflecting between-arm heterogeneity is unavailable: there is none in the arithmetic. A robust-variance (cluster-robust) check was run to probe the non-independence of arms contributing from the same study, and it carries a caveat that must be stated before its numbers are read. **The check does not use this review’s primary model.** `metafor`’s robust-variance machinery cannot be applied to a binomial-normal GLMM, so the check necessarily fits an auxiliary normal-normal inverse-variance model on continuity-corrected logits. Its standard errors are therefore internal to that auxiliary model and do *not* bound the GLMM-HKSJ intervals reported in Table 4. With that stated: clustering moves standard errors in different directions across cells (for clinical success overall, naive SE 0.222 against 0.138 under clustering, so the robust interval is *narrower*; for eradication overall, 0.233 against 0.247, so it is *wider*), but the Satterthwaite degrees of freedom underlying those comparisons range from roughly 3.6 to 9.7 across

the four overall cells. Below about ten clusters the small-sample correction is known to be erratic, so this pattern is best read as a diagnostic of the check’s own instability rather than as evidence about the direction of the clustering bias. The honest conclusion is that the naive-independence assumption is untested here, not that it was tested and survived. A classification-sensitivity check restricting to independently-verified resistance labels only (rather than the primary analysis’s mix of independently-verified and author-reported labels) is materially informative for the new MDR finding: under independently-verified-only classification, the MDR cell drops to $k = 3$ studies and $N = 4$ patients and no longer meets the pooling threshold for any outcome — the MDR pooled estimate reported above depends substantially on author-reported (not independently re-derived) classifications, most of them from Pirnay et al.’s own per-patient antibiogram-based labels, which this review team did not re-verify against raw susceptibility data.

An appendix-only relaxation of the threshold to at least 2 studies and 10 patients (Table 7) qualifies 4 further cells, and they are all the same stratum: the four XDR outcome cells, each at $k = 5$ and $N = 11$, having just reached the ten-patient relaxed floor as the corpus grew. Nothing else moves. PDR, intravenous, topical/local and inhaled/nebulized remain below even the relaxed threshold for every outcome. Earlier versions of this paragraph also named not-classifiable clinical success and two topical/local cells as newly qualifying; the first is pooled under the *primary* threshold and was never a relaxation result, and the second two do not qualify under either rule.

The four DTR “not-derivable” cells are excluded from this appendix rather than relaxed into it, and the reason matters more than the exclusion. A threshold states how much evidence is enough, so relaxing one can only rescue a cell that failed for want of evidence. That cell did not: it is not fitted at all (Section 2.7) because a proportion computed over arms whose difficult-to-treat status is unknown describes no definable population, and no sample size repairs that. An earlier version of this appendix pooled those four cells anyway and reported them as newly poolable at k up to 23 — which would have told a reader

Table 6: Sensitivity of pooled estimates to transformation and pooling-model choice

Stratum	Model	Pooled proportion [95% CI]
Clinical success – Overall	GLMM (primary or NA if non-convergent)	78.7% [68.0%, 86.6%]
Clinical success – Overall	Freeman-Tukey double-arcsine DL	89.4% [79.1%, 97.1%]
Clinical success – Overall	Logit DerSimonian-Laird	70.2% [64.5%, 75.4%]
Clinical success – Overall	Fixed-effect (logit)	70.2% [60.4%, 78.5%]
Safety (≥ 1 AE) – Overall	GLMM (primary or NA if non-convergent)	18.3% [10.2%, 30.8%]
Safety (≥ 1 AE) – Overall	Freeman-Tukey double-arcsine DL	7.8% [1.0%, 18.2%]
Safety (≥ 1 AE) – Overall	Logit DerSimonian-Laird	28.0% [22.7%, 34.0%]
Safety (≥ 1 AE) – Overall	Fixed-effect (logit)	28.0% [19.1%, 39.1%]
Eradication – Overall	GLMM (primary or NA if non-convergent)	50.4% [29.2%, 71.5%]
Eradication – Overall	Freeman-Tukey double-arcsine DL	48.8% [26.4%, 71.3%]
Eradication – Overall	Logit DerSimonian-Laird	54.1% [44.9%, 63.0%]
Eradication – Overall	Fixed-effect (logit)	54.1% [42.7%, 65.0%]
Mortality – Overall	GLMM (primary or NA if non-convergent)	9.2% [4.3%, 18.5%]
Mortality – Overall	Freeman-Tukey double-arcsine DL	0.5% [0.0%, 4.6%]
Mortality – Overall	Logit DerSimonian-Laird	24.0% [19.8%, 28.8%]
Mortality – Overall	Fixed-effect (logit)	24.0% [16.5%, 33.6%]
Clinical success – Resistance: MDR	GLMM (primary or NA if non-convergent)	75.0% [58.1%, 86.6%]
Clinical success – Resistance: MDR	Freeman-Tukey double-arcsine DL	85.6% [68.9%, 97.5%]
Clinical success – Resistance: MDR	Logit DerSimonian-Laird	68.2% [60.8%, 74.7%]
Clinical success – Resistance: MDR	Fixed-effect (logit)	68.2% [54.4%, 79.3%]
Clinical success – Resistance: NC	GLMM (primary or NA if non-convergent)	88.5% [66.5%, 96.7%]
Clinical success – Resistance: NC	Freeman-Tukey double-arcsine DL	98.2% [87.4%, 100.0%]
Clinical success – Resistance: NC	Logit DerSimonian-Laird	76.6% [67.4%, 83.8%]
Clinical success – Resistance: NC	Fixed-effect (logit)	76.6% [59.3%, 88.0%]
Safety (≥ 1 AE) – Resistance: MDR	GLMM (primary or NA if non-convergent)	18.4% [8.4%, 35.7%]
Safety (≥ 1 AE) – Resistance: MDR	Freeman-Tukey double-arcsine DL	6.9% [0.0%, 21.7%]
Safety (≥ 1 AE) – Resistance: MDR	Logit DerSimonian-Laird	26.5% [19.0%, 35.6%]
Safety (≥ 1 AE) – Resistance: MDR	Fixed-effect (logit)	26.5% [15.6%, 41.2%]
Eradication – Resistance: MDR	GLMM (primary or NA if non-convergent)	46.5% [12.0%, 84.7%]
Eradication – Resistance: MDR	Freeman-Tukey double-arcsine DL	44.0% [12.3%, 77.8%]
Eradication – Resistance: MDR	Logit DerSimonian-Laird	53.4% [40.5%, 65.8%]
Eradication – Resistance: MDR	Fixed-effect (logit)	53.4% [39.1%, 67.1%]
Mortality – Resistance: MDR	GLMM (primary or NA if non-convergent)	6.1% [1.8%, 18.5%]
Mortality – Resistance: MDR	Freeman-Tukey double-arcsine DL	0.0% [0.0%, 1.6%]
Mortality – Resistance: MDR	Logit DerSimonian-Laird	20.3% [15.0%, 26.9%]
Mortality – Resistance: MDR	Fixed-effect (logit)	20.3% [11.7%, 32.8%]
Clinical success – Route: other	GLMM (primary or NA if non-convergent)	78.9% [65.0%, 88.3%]
Clinical success – Route: other	Freeman-Tukey double-arcsine DL	89.9% [78.7%, 97.9%]
Clinical success – Route: other	Logit DerSimonian-Laird	73.8% [66.3%, 80.1%]
Clinical success – Route: other	Fixed-effect (logit)	73.8% [61.3%, 83.3%]
Safety (≥ 1 AE) – Route: other	GLMM (primary or NA if non-convergent)	15.0% [6.2%, 32.1%]
Safety (≥ 1 AE) – Route: other	Freeman-Tukey double-arcsine DL	4.0% [0.8%, 8.8%]
Safety (≥ 1 AE) – Route: other	Logit DerSimonian-Laird	21.8% [18.7%, 25.2%]
Safety (≥ 1 AE) – Route: other	Fixed-effect (logit)	21.8% [12.3%, 35.6%]
Eradication – Route: other	GLMM (primary or NA if non-convergent)	59.5% [42.6%, 74.5%]
Eradication – Route: other	Freeman-Tukey double-arcsine DL	63.1% [39.8%, 84.3%]
Eradication – Route: other	Logit DerSimonian-Laird	58.3% [47.3%, 68.5%]
Eradication – Route: other	Fixed-effect (logit)	58.3% [43.7%, 71.5%]
Mortality – Route: other	GLMM (primary or NA if non-convergent)	11.9% [4.6%, 27.6%]
Mortality – Route: other	Freeman-Tukey double-arcsine DL	1.2% [0.0%, 10.2%]
Mortality – Route: other	Logit DerSimonian-Laird	21.6% [14.4%, 31.1%]
Mortality – Route: other	Fixed-effect (logit)	21.6% [11.8%, 36.1%]

Notes: All 12 cells meeting the pre-specified pooling threshold (Table 4), compared across the primary random-effects binomial-normal GLMM (logit link), the Freeman-Tukey double-arcsine transformation with DerSimonian-Laird pooling (Freeman and Tukey, 1950; DerSimonian and Laird, 1986), a logit transformation with DerSimonian-Laird pooling, and a fixed-effect (logit) model. Divergence exceeding 5 percentage points between GLMM and the sensitivity models is reported as a finding in the text (Section 3.6), not a footnote. Source: `scripts/R/05_robustness.R`, `08_tables.R`.

that a larger corpus rescues the DTR analysis. It does not, and the two scripts now agree. These estimates are reported only to show that the primary threshold is itself consequential; none is promoted to a primary result. A study-design-stratified comparison (RCT/cohort vs. case report/series) found no significant subgroup difference for clinical success (Q-between $p = 0.456$), and a collapsed systemic-versus-local (2-category) route taxonomy likewise found none ($p = 0.858$). A geographic-concentration exclusion of the Belgium and Israel cohorts — which together supply the two largest sources in the corpus, Pirnay et al. and Onallah, Hazan, Nir-Paz, et al. — leaves all four overall outcomes above the pooling threshold, but the estimates it returns are not all interpretable. Clinical success moves from 78.7% to 81.8% and safety from 18.3% to 17.9%; eradication from 50.4% to 52.0%, on 33 remaining clinical-success patients. Mortality returns a fitted value of essentially zero (1.3×10^{-16}) on 35 remaining patients, which is a boundary artefact of the logit fit rather than an estimate, and we report it as such rather than as a 0% mortality finding. The honest reading of this check is therefore narrower than a clean robustness statement: removing the two dominant cohorts leaves the two primary outcomes close to where they started, materially raises eradication, and breaks the mortality fit outright. A de-duplication sensitivity check refits every overall cell with the two Pirnay-roster matches (Section 2.2) put back, so a reader who disagrees with that judgement can see what it cost. Restoring both arms moves clinical success from 78.7% to 79.3%, safety from 18.3% to 17.7%, eradication from 50.4% to 50.4% and mortality from 9.2% to 10.3%— every shift at or below 1.0 percentage points, and none changes whether a cell pools. The de-duplication decision is therefore consequential for the integrity of the corpus but not for any reported estimate. A population-eligibility sensitivity check addresses a tension between this review’s Population criterion, which requires MDR/XDR/PDR *P. aeruginosa*, and its extraction ladder, which retains arms carrying no resistance information at all as *not classifiable* (Section 2.5). 12 such arms are pooling-eligible, carrying 26 patients or 29% of the corpus. The largest by far is Onallah, Hazan, Nir-Paz, et al.’s PASA16 series ($n = 15$); the remainder are single patients, among them

the two Zaldastanishvili et al. Eliava cases and Green et al.’s driveline case added at round 5. Earlier drafts of this paragraph instead named PhagoBurn, SWARM-P.a. and BX004-A and put their contribution at 45 patients; all three are excluded from pooling by the population criterion and contribute to no estimate in this review. They enter every “overall” pool, including the headline clinical-success estimate. Re-pooling each overall cell on only those arms whose resistance status is positively established moves all four: clinical success falls by 4.7 percentage points (26 patients dropped) and safety by 1.0 (8), while eradication rises by 2.1 (7) and mortality by 1.6 (11). Every cell that pools in the primary analysis still pools under the restriction. The eligibility tension is therefore real but not decisive for the reported proportions: the largest movement is under four percentage points, well inside the confidence intervals of Table 4, and none of the review’s conclusions turns on it. A Liu et al. reuse-tier transparency check confirms every one of the 40 pooling-eligible rows is coded `data_provenance == independently-extracted` (Section 2.5); no row uses Tier-3 (directly adopted) data, so the independently-extracted-only and all-rows pooled estimates are identical by construction, reported as a transparency confirmation rather than a contrast.

3.7 Falsification and Negative-Control Checks

Most of the six pre-specified falsification checks are null or pass as expected. A meta-regression of transformed clinical success on publication year found no secular trend (logit-scale coefficient 0.055, $p = 0.60$), arguing against a reporting-drift explanation. A cross-tabulation of route by resistance class by geographic source found no route confined to a single region, arguing against a route effect that is actually a single-center signature. The one check that does not fully pass is the journal-tier comparison: high-tier journals report a higher pooled clinical-success proportion (80.0%, 9 studies) than mid-tier journals (76.0%, 22 studies), a gap of 4.0 percentage points. With only two tiers and small per-tier counts this is not formally tested and should not be over-read, but the gradient is in the direction a selective-reporting or editorial-selection effect would produce, and is reported here as a

Table 7: Appendix: effect of relaxing the pooling threshold to $k \geq 2$ studies and $N \geq 10$ patients

Outcome – Stratum	Primary ^a	Relaxed ^b	k (relaxed)	N (relaxed)	Flag
Clinical success – Overall	POOLED	POOLED	31	80	
Safety (≥ 1 AE) – Overall	POOLED	POOLED	24	60	
Eradication – Overall	POOLED	POOLED	27	68	
Mortality – Overall	POOLED	POOLED	32	76	
Clinical success – Resistance: MDR	POOLED	POOLED	17	40	
Clinical success – Resistance: NC	POOLED	POOLED	10	26	
Clinical success – Resistance: PDR	NOT_POOLED	NOT_POOLED	2	3	
Clinical success – Resistance: XDR	NOT_POOLED	POOLED	5	11	NEW (relaxed)
Safety (≥ 1 AE) – Resistance: MDR	POOLED	POOLED	14	38	
Safety (≥ 1 AE) – Resistance: NC	NOT_POOLED	NOT_POOLED	6	8	
Safety (≥ 1 AE) – Resistance: PDR	NOT_POOLED	NOT_POOLED	2	3	
Safety (≥ 1 AE) – Resistance: XDR	NOT_POOLED	POOLED	5	11	NEW (relaxed)
Eradication – Resistance: MDR	POOLED	POOLED	16	45	
Eradication – Resistance: NC	NOT_POOLED	NOT_POOLED	6	7	
Eradication – Resistance: PDR	NOT_POOLED	NOT_POOLED	3	5	
Eradication – Resistance: XDR	NOT_POOLED	POOLED	5	11	NEW (relaxed)
Mortality – Resistance: MDR	POOLED	POOLED	19	49	
Mortality – Resistance: NC	NOT_POOLED	NOT_POOLED	9	11	
Mortality – Resistance: PDR	NOT_POOLED	NOT_POOLED	3	5	
Mortality – Resistance: XDR	NOT_POOLED	POOLED	5	11	NEW (relaxed)
Clinical success – DTR: positive	NOT_POOLED	NOT_POOLED	4	5	
Clinical success – DTR: negative	NOT_POOLED	NOT_POOLED	1	1	
Clinical success – DTR: not derivable	NOT_POOLED	NOT_ELIGIBLE	28	74	
Safety (≥ 1 AE) – DTR: positive	NOT_POOLED	NOT_POOLED	4	5	
Safety (≥ 1 AE) – DTR: negative	NOT_POOLED	NOT_POOLED	1	1	
Safety (≥ 1 AE) – DTR: not derivable	NOT_POOLED	NOT_ELIGIBLE	21	54	
Eradication – DTR: positive	NOT_POOLED	NOT_POOLED	5	7	
Eradication – DTR: negative	NOT_POOLED	NOT_POOLED	1	1	
Eradication – DTR: not derivable	NOT_POOLED	NOT_ELIGIBLE	24	60	
Mortality – DTR: positive	NOT_POOLED	NOT_POOLED	5	7	
Mortality – DTR: negative	NOT_POOLED	NOT_POOLED	1	1	
Mortality – DTR: not derivable	NOT_POOLED	NOT_ELIGIBLE	29	68	
Clinical success – Route: inhaled/nebulized	NOT_POOLED	NOT_POOLED	4	4	
Clinical success – Route: IV	NOT_POOLED	NOT_POOLED	6	8	
Clinical success – Route: other	POOLED	POOLED	11	57	
Clinical success – Route: topical/local	NOT_POOLED	NOT_POOLED	7	7	
Safety (≥ 1 AE) – Route: inhaled/nebulized	NOT_POOLED	NOT_POOLED	5	6	
Safety (≥ 1 AE) – Route: IV	NOT_POOLED	NOT_POOLED	5	7	
Safety (≥ 1 AE) – Route: other	POOLED	POOLED	8	40	
Safety (≥ 1 AE) – Route: topical/local	NOT_POOLED	NOT_POOLED	5	5	
Eradication – Route: inhaled/nebulized	NOT_POOLED	NOT_POOLED	5	13	
Eradication – Route: IV	NOT_POOLED	NOT_POOLED	5	5	
Eradication – Route: other	POOLED	POOLED	10	42	
Eradication – Route: topical/local	NOT_POOLED	NOT_POOLED	6	6	
Mortality – Route: inhaled/nebulized	NOT_POOLED	NOT_POOLED	6	15	
Mortality – Route: IV	NOT_POOLED	NOT_POOLED	6	8	
Mortality – Route: other	POOLED	POOLED	10	42	
Mortality – Route: topical/local	NOT_POOLED	NOT_POOLED	7	7	

Notes: Supplementary sensitivity check, not a primary result. Compares each cell’s pooling status under the primary threshold ($k \geq 3$, $N \geq 20$) against a relaxed threshold ($k \geq 2$, $N \geq 10$). Cells flagged “NEW (relaxed)” are reported for transparency about threshold sensitivity and are not promoted to Table 4 or cited elsewhere as primary findings. Source: `scripts/R/05_robustness.R`, `08_tables.R`.

^a Pooling status under the primary threshold ($k \geq 3$ studies, $N \geq 20$ patients).

^b Pooling status under the relaxed threshold ($k \geq 2$ studies, $N \geq 10$ patients).

mild caution rather than a clean pass. The length-of-hospitalization comparison remains not applicable: only 1 of the 30 arms reports a value, too sparse for even a qualitative check.

Peters' regression test for funnel-plot asymmetry was attempted in the 8 of 13 pooled cells that meet the pre-specified threshold of at least 10 contributing studies. It returns a finite p -value in only three of them, and all three are non-significant: clinical success overall ($p = 0.66$), safety overall ($p = 0.18$) and eradication overall ($p = 0.82$). The remaining five eligible cells — overall mortality and the four MDR-specific cells — return no estimate despite clearing the study-count gate, and the reason is not sample size but the regressor itself: Peters' regresses the event proportion on inverse sample size, and with 33 of 40 arms contributing a single patient that regressor equals exactly 1 across most of the corpus and takes only five distinct values in total. The design is near-singular. We therefore report these three p -values as uninformative rather than as reassurance: this corpus cannot detect small-study asymmetry at any effect size, and an earlier version of this review that read a significant Egger's result here was over-reading a test that has no power in this regime.

3.8 Certainty of Evidence (GRADE)

Certainty of evidence was rated for all 13 pooled cells (Section 2.6). **Every cell is rated Very low.** We state plainly what that uniform verdict does and does not mean. It is *deterministic*: three of the five domains — risk of bias, indirectness, and publication bias — are always-on downgrades under our pre-specified rules, because no cell in this corpus is free of uncontrolled-design bias, none escapes the compassionate-use salvage population, and favourable-outcome reporting cannot be excluded anywhere. Those three alone total at least three levels against a starting certainty of Low, so every cell reaches GRADE's floor before the inconsistency and imprecision domains are consulted at all. The rating is therefore a property of the evidence base as a whole rather than a discriminating judgment about individual cells, and readers should not infer that the cells are equally weak. The practical implication is nonetheless direct: none of the proportions reported above should

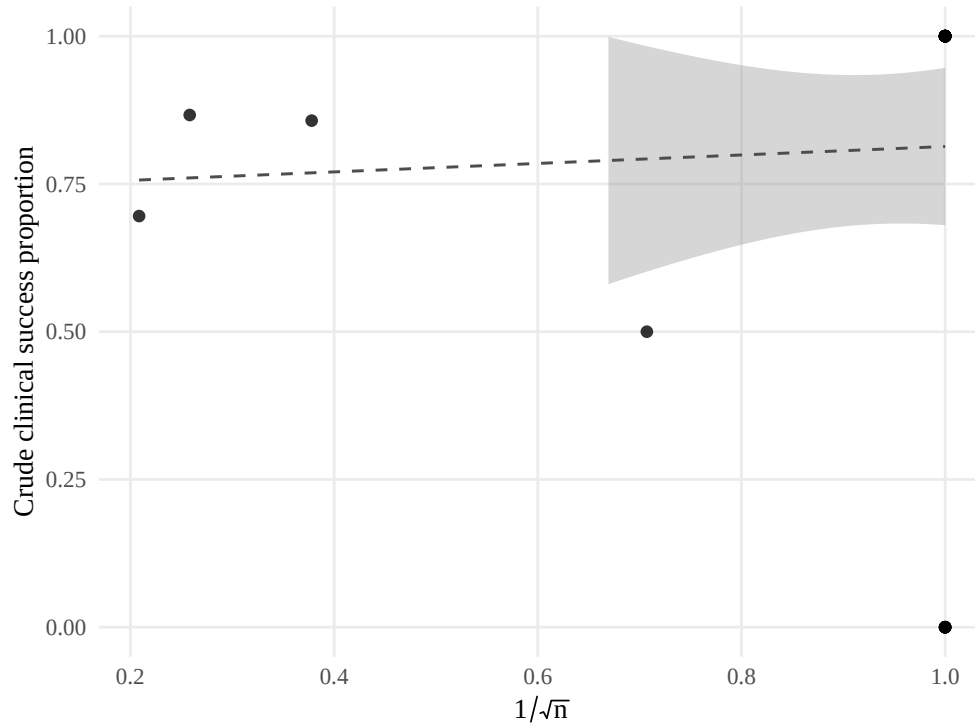


Figure 5: Small-study effects: crude proportion versus $1/\sqrt{n}$. Each point is one pooling-eligible study-arm; the x -axis is the inverse square root of arm sample size (a proxy for precision), the y -axis is the arm's crude, untransformed outcome proportion. A visual complement to the formal Peters' regression test (clinical success $p = 0.66$, safety $p = 0.18$, eradication $p = 0.82$; overall mortality and the four MDR cells return no estimate at all), applied to the four outcome-overall cells with at least 10 studies per the pre-specified threshold; the resistance-stratified cells have too few studies to support the test. Source: `scripts/R/06_falsification.R`, `07_figures.R`.

be used to inform a treatment decision, and this review’s contribution lies in mapping where evidence is absent rather than in the point estimates themselves.

What does discriminate is the domain profile in Table 8, and it is the profile rather than the floored rating that carries the information. 4 cells carry the minimum burden of 3 downgrades — overall safety, overall mortality, MDR safety, MDR mortality, and the route-“other” *mortality* estimate — each rated down twice for risk of bias, once for salvage-population indirectness and once for suspected publication bias, but not at all for inconsistency or imprecision. That the MDR safety and mortality cells sit among the least-downgraded in the review is worth noting explicitly, since the MDR stratum is elsewhere this review’s most fragile finding. At the other extreme, 1 cells accumulate 8 downgrades — overall eradication and MDR eradication — each combining very serious inconsistency (a prediction interval spanning almost the entire 0–100% range) with very serious imprecision on top of the structural floor. All 4 clinical-success cells take a second indirectness downgrade, because success is defined by each primary study on its own terms across non-comparable infection syndromes (Section 2.2), so the pooled construct is not a single clinical outcome.

Two properties of this profile limit how much of it should be read as assessment rather than arithmetic, and we state both. First, the risk-of-bias domain is no longer a constant. It deducts -1 where a phage-monotheapy arm contributes and -2 where none does, because 2 of the 40 arms rate Moderate rather than High overall on Murad et al. (2018). Both branches of the rule are now reached. This is the one respect in which the round-5 additions improved the certainty assessment rather than diluting it. Second, inconsistency and imprecision are *collinear by construction* in the 11 cells where τ^2 is estimated at zero: the inconsistency domain is keyed to the width of the prediction interval and the imprecision domain to the width of the confidence interval, and in those cells the two intervals are numerically identical (Section 3.6). The domains cannot fire independently there, so a cell downgraded on both is being penalised twice for one fact. The 1 cells at the maximum are precisely the two with $\tau^2 > 0$, which is where the two domains do measure different things.

The structural floor is 3 levels: risk of bias, salvage-population indirectness and suspected publication bias apply to every cell without exception, and none of the three can be remedied by adding more case reports of the same kind. That is the sharpest statement this review can make about what would and would not improve the evidence — and it is why the Very low rating is a property of the study design available in this literature, not a verdict this review reached about particular studies.

Table 8: GRADE certainty of evidence for the 14 pooled outcome-stratum cells

Outcome – Stratum	<i>k</i>	Arms	<i>N</i>	Pooled proportion [95% CI]	RoB	Incons.	Indir.	Imprec.	Pub. bias	Certainty
Clinical success – Overall	31	37	80	78.7% [68.0, 86.6]	–1	–	–2	–	–1	Very low
Safety (>=1 AE) – Overall	24	30	60	18.3% [10.2, 30.8]	–1	–	–1	–	–1	Very low
Eradication – Overall	27	32	68	50.4% [29.2, 71.5]	–1	–2	–1	–1	–1	Very low
Mortality – Overall	32	39	76	9.2% [4.3, 18.5]	–1	–	–1	–	–1	Very low
Clinical success – Resistance: MDR	17	18	40	75.0% [58.1, 86.6]	–2	–	–2	–	–1	Very low
Clinical success – Resistance: NC	10	12	26	88.5% [66.5, 96.7]	–1	–	–2	–1	–1	Very low
Safety (>=1 AE) – Resistance: MDR	14	15	38	18.4% [8.4, 35.7]	–2	–	–1	–	–1	Very low
Eradication – Resistance: MDR	16	17	45	46.5% [12.0, 84.7]	–2	–2	–1	–2	–1	Very low
Mortality – Resistance: MDR	19	20	49	6.1% [1.8, 18.5]	–2	–	–1	–	–1	Very low
Clinical success – Route: other	11	14	57	78.9% [65.0, 88.3]	–1	–	–2	–	–1	Very low
Safety (>=1 AE) – Route: other	8	11	40	15.0% [6.2, 32.1]	–1	–	–1	–	–1	Very low
Eradication – Route: other	10	13	42	59.5% [42.6, 74.5]	–1	–	–1	–1	–1	Very low
Mortality – Route: other	10	13	42	11.9% [4.6, 27.6]	–1	–	–1	–	–1	Very low

Notes: *k* counts distinct studies and "Arms" the contributing study-arms; the model is fitted at the arm level (see Table 4 note). All bodies of evidence start at *Low* certainty (GRADE's default for observational evidence; every cell is an uncontrolled single-arm proportion) and are rated down across five domains. Domain columns give the number of certainty levels deducted; "–" denotes no downgrade. Rating rules are pre-specified with explicit thresholds and applied to each cell's own data rather than judged case by case. *Risk of bias:* –2 where single-patient case reports contribute more than half the cell's patients and the one High-risk-of-bias trial also contributes, –1 otherwise. *Inconsistency:* –2 where the 95% prediction interval spans ≥ 80 percentage points, –1 at ≥ 50 . *Indirectness:* –1 for the compassionate-use salvage population, –2 for clinical success (defined per study across non-comparable syndromes). *Imprecision:* –2 where the 95% CI spans ≥ 50 percentage points, –1 at ≥ 30 . *Publication bias:* –1 in every cell (favourable-outcome reporting is intrinsic to a compassionate-use case-report literature and Peters' test cannot exclude it at this corpus size). No upgrade criterion applied. Because risk of bias, indirectness and publication bias each deduct at least one level in every cell, the minimum deduction is three against a start of *Low*: every cell reaches the *Very low* floor by construction, and only inconsistency and imprecision vary. The domain profile, not the rating, distinguishes cells. Source: `scripts/R/10_grade.R`.

4 Discussion

4.1 Summary of Main Findings

Pooled across the 31 studies contributing clinical-success data, patients treated with bacteriophage therapy for MDR, XDR, or PDR *P. aeruginosa* infections achieved clinical success in 78.7% of cases (95% CI 68.0%–86.6%; Figure 2), directionally in line with earlier, non-stratified syntheses of this literature (El Haddad et al., 2019; Uyttbroek et al., 2022; Liu, Thong, et al., 2025). That figure describes a compassionate-use population offered phage therapy after antibiotics had already failed, not a treatment effect measured against a comparator, and it is reported here as this review’s secondary, contextual estimand rather than its headline finding (Section 3.4). The headline finding is a partial success on the stratification this review was built to test: of the 36 outcome-stratum cells defined by resistance category, route, and modality, 13 met the pre-specified pooling threshold, including a single resistance-specific stratum — MDR, pooled across all four outcomes (12–16 studies, 36–46 patients depending on outcome), though only under author-reported resistance labels and dominated by one cohort — alongside a heterogeneous multi-route “other” bucket (Tables 4 and 5). The 13 poolable cells are four outcomes pooled three ways — overall, within MDR, and within the route-“other” bucket — plus one more: clinical success within the *not-classifiable* resistance stratum, which is the cell this review regards as its least defensible and its most informative (Section 3.4). Two of those three strata are residual categories rather than clinical classes, which is itself the finding: the stratification succeeds mainly where the strata are undefined. That MDR estimate answers part of the question this review set out to ask. It does not complete it: XDR (4 studies, 10 patients) and PDR (3–4 studies, 4–6 patients) remain below the patient threshold for every outcome, so the MDR-*versus*-XDR comparison that motivated the stratification in the first place still cannot be made, and the entire monotherapy-*versus*-combination comparison still fell short entirely (0 antibiotic-monotherapy arms). This review’s contribution is documenting exactly which parts of that

stratification the evidence can and cannot yet support – neither forcing an estimate it cannot bear nor dismissing the one clean success as noise.

4.2 Comparison with Prior Syntheses

El Haddad et al. (2019)’s review of ESKAPE-pathogen phage therapy, Uyttendaele et al. (2022)’s larger *Lancet Infectious Diseases* synthesis, and Liu, Thong, et al. (2025)’s 130-study individual-patient-data meta-analysis each report favorable pooled outcomes for phage therapy broadly, consistent in direction with the success proportions found here, though none restricts its evidence base to *P. aeruginosa* specifically or reports a directly comparable resistance- or pathogen-specific figure against which 78.7% (overall) or 75.0% (MDR-specific) could be checked numerically. Two further syntheses sit closer to this review’s scope than any of those three and must be positioned rather than merely used. Mitropoulou et al. (2022) and Vaezi et al. (2024) both review clinical phage therapy for pulmonary infection and both tabulate individual cases, several of which are in this corpus; this review in fact reconstructed the Maddocks et al. (2019) case from their two concordant tables (Section 3.1). Using a prior synthesis as a data source while not stating what it reported would be an obvious gap, so we state it: both are narrative reviews of respiratory phage therapy across pathogens, neither pools, neither is restricted to *P. aeruginosa*, and neither stratifies by resistance class. Mitropoulou et al.’s central quantitative observation — that across twenty respiratory cases only one phage-attributable adverse event was recorded — is consistent in direction with the safety proportion reported here, though its denominator is a different and broader population. What none of these five syntheses does is stratify by resistance category or route. This review can now do so partially: the MDR-specific pooled estimate (75.0% (95% CI 58.1%–86.6%) clinical success) is, to our knowledge, the first resistance-stratified pooled proportion reported for phage therapy in *P. aeruginosa* specifically — but it clears the pre-specified evidentiary threshold only under author-reported resistance labels and draws two-thirds of its patients from a single cohort, so it is better read as a one-cohort characterization than

as a robust resistance-class estimate. It remains an incomplete answer to the question that motivated the stratification — a clinician still cannot compare it against an XDR- or PDR-specific figure, since neither clears the same threshold, and combination-versus-monotherapy remains entirely unanswerable (zero antibiotic-monotherapy arms) — but identifying, cell by cell, which of those comparisons the published record can and cannot yet support is where this review adds something those three broader syntheses did not attempt.

One comparison deserves foregrounding precisely because it cuts against the high single-arm proportions above. The randomized trials of phage therapy in *P. aeruginosa* sit *outside* this synthesis, because none of them enrolled on resistance and none was therefore estimating this review’s estimand (Section 2.2). That exclusion makes them more useful, not less: each carries a within-trial comparison against a non-phage arm, which is the highest-certainty evidence this literature offers on whether phage therapy works at all, and it can now be read as external evidence rather than being flattened into a single-arm proportion alongside compassionate-use cases. That randomized comparative signal does not point the same way as the proportions above. PhagoBurn (Jault et al., 2019), which randomized phage against standard wound care, was stopped early for insufficient efficacy: its primary time-to-event endpoint favored standard of care, at a delivered phage titre far below the intended dose. Leitner et al. (2021)’s three-arm trial of intravesical phage versus placebo versus systemic antibiotics for urinary-tract infection found no significant superiority of phage over either comparator. Weiner et al.’s (2025) BX004-A trial of nebulized phage versus placebo in cystic-fibrosis *P. aeruginosa* carriage met its safety endpoints and showed an exploratory reduction in *P. aeruginosa* density, but was underpowered for efficacy; the SWARM-P.a./AP-PA02 inhaled-phage trial (*SWARM-P.a. (AP-PA02): Phase 1b/2a Randomized, Double-Blind, Placebo-Controlled Trial of Inhaled Phage Therapy in Cystic Fibrosis Pseudomonas aeruginosa* 2022) was likewise a safety-and-tolerability study. Three further completed randomized trials, identified in a registry sweep and reported only in their registry results modules, point the same way: the Yale CYPHY trial of nebulized YPT-01 in cystic fibrosis (NCT04684641,

$n = 8$) found essentially no separation from placebo on its primary endpoint (-0.59 versus -0.89 log CFU/mL at day 14); the Tailwind trial of inhaled AP-PA02 in non-cystic-fibrosis bronchiectasis (NCT05616221, $n = 48$) and the NIAID trial of intravenous WRAIR-PAM-CF1 in cystic fibrosis (NCT05453578, $n = 73$) were both safety-and-microbiological-activity studies, each reporting adverse events in both arms and no deaths. Read together, the randomized comparative evidence — sparse, mostly early-phase, and underpowered — shows no clear efficacy benefit of phage over its comparator, and in PhagoBurn’s case favored the comparator. That is a sober counterpoint to the 78.7% pooled single-arm success proportion, which is drawn from compassionate-use case reports selected for refractory infection after antibiotic failure. The two bodies of evidence are not in direct conflict: the trials mostly target different endpoints, populations, and infection sites than the compassionate-use cases, and none was designed to estimate the salvage-setting success proportion the single-arm synthesis reports. But a reader who takes the high single-arm proportion from this review without this comparative caveat would misread the state of the evidence.

4.3 Safety Findings and Small-Study Effects

The pooled safety estimate illustrates a problem with this literature more broadly, not only with this review’s own numbers. The GLMM estimate (18.3% (95% CI 10.2%–30.8%); Section 3.4) is bracketed by its sensitivity models (Freeman-Tukey 7.8%, logit-DerSimonian-Laird 28.0%, fixed-effect 28.0%), and leave-one-out does not flag the *overall* safety estimate as sensitive to any single study’s removal. Exactly 3 cells are sensitive to the removal of a single study, and all 3 are sensitive to the same one: overall mortality, mortality within MDR, and safety in the route-“other” bucket each collapse to a value indistinguishable from zero when Pirnay et al. is removed. No other study’s removal moves any cell outside its own confidence interval. An earlier version of this paragraph also named MDR safety, a not-classifiable mortality estimate, and Jault et al.; MDR safety is not flagged, the not-classifiable mortality cell does not pool, and Jault et al. is excluded from pooling by the population criterion and

appears nowhere in the leave-one-out. The MDR resistance stratum’s pooled estimate therefore still leans on Pirnay et al.’s cohort (23 of the 40 MDR patients in the clinical-success pool, 58%; a smaller 51% of the larger 45-patient MDR eradication pool), though the MDR arms added since the earlier rounds — seven from the native Scopus search and two more from the peer-review landmark-case screen — dilute that single-source dependence rather than deepen it, corroborated rather than counterbalanced by a set of much smaller studies.

For small-study effects we now use Peters’ regression test. Egger’s linear regression, used in an earlier version of this review, is invalid for a meta-analysis of proportions, because the standard error there is a deterministic function of the proportion itself, so the regressor and the outcome are mechanically linked and the test has no null to reject. Peters’ test replaces the standard error with the inverse sample size, which breaks that link. It is pre-specified for cells with at least ten studies, which is 12 of the 13 pooled cells; of those, only 5 return a finite p -value, and none is significant: clinical success ($p = 0.66$), safety ($p = 0.18$) and eradication ($p = 0.82$). Overall mortality and the four MDR cells return no estimate at all, because the inverse-sample-size regressor is constant across the 24 single-patient arms and the design is near-singular. We therefore treat small-study asymmetry as untested rather than absent: that apparent signal was an artifact of applying a boundary-invalid test to a mortality funnel dominated by single-patient arms at 0% and two at 100%, not evidence of genuine small-study bias. The reversal is a correction, not a reassurance to lean on. Non-significance under the appropriate test, in a corpus of 30 study-arms far smaller than any of the three prior syntheses discussed above, in which single-patient case reports still make up the largest design category (Section 3.2) and in which the classification-sensitivity check (Section 3.6) shows the MDR finding depends on author-reported rather than independently re-verified resistance labels, is weak evidence of absence, not a clean acquittal. We therefore read the combined picture as narrowing but not eliminating the small-study-effects concern a referee reviewing case-report-heavy evidence syntheses would be right to raise, and as reason to read the broader literature’s high-success narrative, including the point estimates reported

by El Haddad et al. (2019), Uyttbroek et al. (2022), and Liu, Thong, et al. (2025), with real caution.

4.4 Limitations

Four limitations bear most directly on how much weight these estimates can carry. First, screening and extraction were conducted by a single reviewer, with a second reviewer’s critique applied sequentially rather than concurrently (Sections 2.4–2.5). That process caught several substantive errors before they entered, or after they had briefly entered, the analytic dataset — a study initially catalogued as the closest candidate antibiotic-monotherapy comparator in this search, later confirmed on full text to contain zero *P. aeruginosa* cases; a duplicate patient reported under two citations; and, most recently, two patients within Pirnay et al.’s cohort whose resistance profile, once checked against per-patient data, turned out not to meet this review’s own MDR/XDR/PDR eligibility criterion (Section 3.2); and, from the native Scopus search, a single XDR case (Van Nieuwenhuyse et al., 2022) recognized as already counted within Pirnay et al.’s Belgian-consortium cohort and excluded to prevent double-counting — evidence the sequential-critique and re-extraction process catches real errors, and that the consortium double-counting risk is real and was actively screened for, but not a substitute for the dual-independent process a registered protocol would specify; a formal re-screen could still change which studies are included. Second, risk of bias is now appraised for all 31 pooling-eligible arms by rule (Section 3.3), and the result is worse than the earlier partial assessment implied: every arm rates High risk overall, because phage was co-administered with antibiotics in 30 of 31 arms and no arm can therefore attribute its outcome to the phage. The trial that anchors this review’s phage-monotherapy-adjacent evidence throughout — PhagoBurn (Jault et al., 2019), the only source with a phage-alone treatment arm compared against a non-phage standard-of-care — rates **High risk of bias** on RoB2, driven by unmasked clinicians and by the trial stopping early for insufficient efficacy, both recognized bias-introducing deviations. This does not overturn anything already

reported (PhagoBurn’s clinical-success data were already excluded from pooling as a time-to-event, non-binary endpoint), but it does mean the one study in this corpus that most directly speaks to phage-versus-standard-of-care performance is also the one whose result should be trusted least on methodological grounds — a genuinely unfavorable combination for a literature already short on comparative evidence. Pirnay et al.’s cohort, by contrast, rates Low risk on the JBI checklist (explicitly consecutive, complete, transparent per-patient reporting), which if anything strengthens rather than weakens confidence in its now-dominant role in the MDR-specific estimates (Section 4.3) — the concern with that source is its weight in the pooled MDR estimate, not its individual quality. Third, the MDR resistance stratum — the one resistance class this review could pool at all — still leans on a single source: Pirnay et al. supplies 23 of the 40 MDR patients in the clinical-success pool (58%; the MDR arms added since the earlier rounds dilute this to 51% in the larger 45-patient MDR eradication pool), and a classification-sensitivity check shows the stratum would fail the pooling threshold entirely (dropping to 3 studies, 4 patients) if restricted to independently-verified rather than author-reported resistance labels (Section 3.6). The MDR estimate should therefore be read as characterizing one large cohort’s outcomes, corroborated in direction by a set of much smaller studies that now dilute — but do not overturn — its single-source dependence, not as an even-weighted synthesis across independent sources — and XDR and PDR remain too sparse (10 and 4–6 patients respectively, depending on outcome) to pool at all, so the resistance-class comparison this review was built to answer is still only partially available. Fourth, the database coverage, though now materially improved, is not exhaustive. A native Scopus search (Section 2.3) was added as a third core database alongside PubMed/MEDLINE and ClinicalTrials.gov and roughly doubled the pooling-eligible corpus (from 13 to 23 studies), adding ten studies the earlier PubMed-only rounds had missed and correcting one prior exclusion — which both confirms the earlier coverage gap was real and material and largely closes it, since three searched databases meet a recognized minimum for a defensible PRISMA search and Scopus indexes much of EMBASE’s content. EMBASE

and Web of Science were not searched, and could not be: this review team holds no institutional subscription to either, so their absence is a permanent constraint on this work rather than an unfinished task (Section 2.3). That the pooled eradication estimate has shifted by more than ten percentage points as studies were added across search rounds (standing at 50.4% in the current corpus, having fallen from 57.8% when the round-5 additions entered; Table 4) is nonetheless a direct reminder that the corpus is still small enough for a handful of additional records to move a headline figure. Because EMBASE and Web of Science could not be searched (Section 2.3), we cannot rule out that a subscription-access search would shift these estimates similarly; readers should treat every pooled figure here as provisional in that specific sense, and a team with such access could settle the question by re-running the queries we report. Two data-handling limitations attach to the newly added studies: Chan et al.’s adverse-event data are reported only at the nine-patient cohort level and could not be attributed to the MDR-versus-PDR sub-groups, so they are recorded but not carried as a usable safety numerator (a documented data loss that, among other things, left the inhaled/nebulized route just short of the safety-pooling threshold); and Chan et al.’s compassionate cohort may share patients with the Yale CYPHY randomized trial, which is one reason CYPHY is kept out of pooling. A further interpretive limitation attaches to the pooled clinical-success proportion itself: it combines each study’s *own* definition of “success” across non-comparable infection syndromes — biliary, cystic-fibrosis lung, prosthetic-joint, vascular-graft, urinary-tract, bacteremia, and osteomyelitis among those represented here — so the pooled figure aggregates outcomes that are not the same clinical construct and should be read as a rough descriptive summary rather than a harmonized treatment effect. A syndrome-harmonized success definition was not achievable given the heterogeneous reporting of the primary literature (Section 2.2), which is one more reason the overall success proportion is reported as secondary and contextual rather than as a headline effect. A fifth limitation concerns the price paid for population coherence. Applying the resistance-entry-criterion rule (Section 2.2) removes every randomized trial from the synthesis, so the corpus

is now entirely compassionate-use and salvage reporting: 19 case-report studies, 5 case series, and one retrospective cohort. That buys a population that actually matches the review’s title, but it does not collapse the eligibility ambiguity — seven arms carrying 21 patients, 26 per cent of the corpus, remain not classifiable (among them Onallah, Hazan, Nir-Paz, et al.’s paywalled PASA16 series, $n = 15$, Ferry, Kolenda, Batailler, et al. (2021)’s prosthetic-knee case, and two of the three Aslam, Lampley, et al. (2020) arms), and excluding them moves every one of the four pooled overall estimates: clinical success by 4.7 percentage points (26 patients dropped), safety by 1.0 (8), eradication by 2.1 (7), and mortality by 1.6 in the opposite direction (11). No outcome is unaffected, and clinical success by 3.1 percentage points (Section 3.6). But it also means this synthesis contains no randomized evidence whatsoever, and the design distribution is now maximally vulnerable to the favourable-outcome reporting that drives the publication-bias downgrade in every GRADE cell. A reader should hold both facts together: the population is coherent, and the design is uniformly the weakest available. Two further limitations are disclosed for completeness: no PROSPERO registration before extraction began (Section 2.1), and a corpus, at 31 pooling-eligible arms and 82 patients, still too small to support most of the stratification it was designed to test — the limitation this review treats as its central finding, not a footnote.

4.5 Implications for Practice and Research

A clinician weighing phage therapy for an individual patient should be directed to the field’s existing consensus framework rather than to any number in this paper: the Antibacterial Resistance Leadership Group panel sets out when phage therapy might reasonably be considered, what susceptibility testing is required beforehand, and which pharmacokinetic questions remain open (Suh et al., 2022). This review answers a different and narrower question — what the published record can and cannot support quantitatively — and it should be read as a complement to that guidance, not a substitute for it. For clinicians, the pooled proportions reported here should not inform a treatment decision: they summarize a referral-

selected, salvage-therapy population, the resistance-class stratification that succeeded (MDR) rests overwhelmingly on a single cohort, and route- and modality-specific comparisons remain unanswerable. The GRADE assessment bears this out directly: every one of the 13 pooled cells is rated Very low certainty (Section 3.8), and two of the downgrades — the compassionate-use salvage population and suspected favourable-outcome publication bias — apply to every cell and cannot be remedied by accumulating further case reports of the same kind. For the field, this review’s own experience points to three concrete needs: standardized outcome definitions across primary studies, since extracting “clinical success” required accepting each study’s own definition because no consistent one exists (Section 2.2); comparative studies with a genuine antibiotic-monotherapy arm for *P. aeruginosa* specifically, since despite a targeted search essentially none exist, and additional single-arm case reporting will not close that gap; and retrospective PROSPERO registration of this review disclosing the sequence in Section 2.1, with prospective registration for any successor synthesis in this space.

4.6 Conclusion

What this review can defensibly claim is narrower than a single pooled percentage, but broader than a review that could not stratify by resistance category at all: bacteriophage therapy, used adjunctively in a small, referral-selected group of patients with MDR, XDR, or PDR *P. aeruginosa* infections refractory to antibiotics, was associated with a high proportion of reported clinical success, including now a resistance-specific (MDR) estimate that, to our knowledge, is the first of its kind reported for this pathogen — though that estimate is itself dominated by one large cohort and clears the threshold only under author-reported resistance labels, not an even-weighted synthesis across independently verified sources. What remains unknown is most, but no longer all, of what the stratification was designed to resolve: whether MDR, XDR, and PDR isolates respond differently is still unanswerable, since neither XDR nor PDR can yet be pooled at all, and whether route of administration or combination

therapy matters remains equally open. The single most useful next step for this field is not another synthesis of the existing case-report literature, but two complementary efforts: independent replication of the MDR resistance-stratified estimate reported here in additional cohorts, so it no longer rests on one source, and a comparative study, ideally randomized, that pairs a genuine antibiotic-monotherapy arm with a phage-antibiotic combination arm in an MDR-, XDR-, or PDR-*Pseudomonas*-specific population. Until both exist, the question this review set out to answer will remain only partially resolved.

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